Context length and readout parameterisation select the emergence exponent of in-context skills in an additive model

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Abstract

We study how a feature-learning model acquires *skills* that are presented in context. Targets are additive models $y = \sum_p c_p\, \sigma(\langle v_p, x\rangle)$ with mean-zero task coefficients c_p, and a student with learned features and a linear-in-label attention readout must recover the directions v_p by online SGD from prompts of length N. For a single skill of information exponent k^* we obtain the population loss in closed form. As $N \to \infty$ it is the in-weight loss with the link squared (effective exponent $2k^*$, a fact implicit in one-step analyses that we make dynamical). At finite N the loss acquires an *alignment-dependent* variance term $\Gamma^2 V(m)/N$, and the way the readout Γ is parameterised decides what this term does: a readout pinned at order one makes the unaligned state drift-stable below m^*, so that more prompts cannot substitute for context length; a free scalar readout shrinks to a Wiener factor: for the population flow at fixed context length the exponent becomes $4k^*$ with escape time $\propto 1/N$ (Theorem B), while at accessible d online SGD reproduces the flow's intermediate slowdown, set jointly by context length and the readout's initialisation and learning rate; a readout tied to the feature norm removes the N-dependence as long as its initial norm is below an explicit threshold (proved, with the N-dependence bounded by $k^*\rho_0/(N m_0^{2k^*-2})$). A parameter-free population ODE reproduces the exponents measured by online SGD at $d = 32$ for three readout protocols and four context lengths (all 12 cells within 0.5 by a censored fit), the escape times at $d = 64$ and $d = 128$ including the predicted flip of a context length from escape to trap, and, at $k^* = 3$, the trap threshold and an exponent near 6 (two pre-registered precision bands were missed and are reported). With many skills the dynamics decouple with emergence times $\propto 1/\pi_p$; additive compositions of learned skills follow by linearity. In small softmax transformers no parameter whose scale we froze reproduces the trap or the rescue (two pre-registered tests, both killed), and prompts per step partly substitute for context length, which the population loss cannot produce; the mechanism is open.

1 Introduction

Theories of feature learning explain *when* a network acquires a direction v in input space in terms of the information exponent k^* of the link function [2]: online SGD needs d^{k^*-1} samples ($d \log d$ for $k^* = 2$). Theories of emergence and scaling laws [12, 13, 17] then sum many such acquisitions over a power-law dictionary of skills. Both lines treat the skill as something learned *in the weights* from a fixed task. Large models, however, acquire much of their behaviour as *in-context* skills: the pretraining data presents a skill through prompts whose labels are generated by a task-dependent coefficient, and the network must learn features that make the skill recoverable from a context of length N. Recent analyses of in-context learning with learned features [7, 9, 14, 16] treat the context length as an *inference-time* resource: it controls how well a trained model estimates the

task, and in one-step analyses of pretraining it enters only through concentration, so that more prompts can substitute for longer prompts [16, Remark 3].

We ask a different question: *does the context length, together with how the readout is parameterised, change which exponent governs the acquisition of a skill during pretraining?* We answer it in the smallest setting in which the question is sharp—one or several skills with $k^* = 2$ (even links, where in-weight learning is easy), a student with features on the sphere and a linear-in-label readout, online SGD—and then test the predictions outside the solvable model.

Contributions. (0, setup) **Closed form.** The population in-context loss of a single feature is $L_A(m) = 1 - 2\Gamma g(m)^2 + \Gamma^2 g(1)\big[(1 - \frac{1}{N})g(m)^2 + V(m)/N\big]$, where m is the overlap with the skill direction, $g(m) = \mathbb{E}[\sigma(\langle v, x\rangle)\sigma(\langle w, x\rangle)]$ and $V(m) = \mathbb{E}[\sigma^2\sigma^2]$ (Prop. 1). As $N \to \infty$ it is the in-weight loss with $g \mapsto g^2$: the effective exponent is $2k^*$. The m^{2k^*-1} gradient is inside the one-step analysis of Oko et al. [16, Lemma 21] (and depends on parameterisation and initialisation, cf. Nishikawa et al. [14]); we make the statement dynamical and verify it by online SGD, including the $k^* = 2$ case excluded from Ren et al. [17]. (1) **The finite-context term and the readout** (Props. 2–3). $\Gamma^2 V(m)/N$ grows with alignment. With Γ pinned at order one the origin is drift-stable below m^*, $m^{*2} = 4\Gamma/\big(2N - (N+5)\Gamma\big)$; under SGD the threshold is soft, and more prompts per step do not remove it (Fig. 3). With a free scalar readout, Γ moves towards the Wiener value $\Gamma^*(m) = g^2/\big(g(1)[(1 - \frac{1}{N})g^2 + V/N]\big)$ and multiplies the feature gradient: the effective exponent rises from $2k^*$ towards $4k^*$ as the context statistic becomes noise-dominated; in the adiabatic limit the escape time scales as $1/N$ there—the multiplicative trade-off of one-step analyses [16, Remark 3] reappears as a consequence of shrinkage—and at finite readout learning rates the ODE gives an intermediate, parameter-dependent N-dependence that we test prospectively. With the readout tied to the feature norm (the 2-homogeneous student of Ren et al. [17]) the directional dynamics are norm-independent (their Lemma B.1) and the N-dependence disappears. The population ODE predicts the exponents measured by SGD at $d = 32$ (Fig. 1). (2) **Many skills and composition.** With a power-law dictionary the dynamics decouple with $T_p \propto 1/\pi_p$ (measured slope 1.44–1.53 for $\pi_p \propto p^{-1.5}$) against 0.75–0.89 for two in-weight baselines, a corollary of (1); additive compositions of learned skills emerge with the slower skill, by linearity. (3) **Transformers.** In a two-layer softmax transformer with small initial alignment ($d = 256$, width 32) a context of 16 with 16× the prompts emerges ≈ 3× later than a context of 256 at equal tokens per step; with large initial alignment ($d = 32$, width 256) the ordering reverses. Neither pre-registered extreme (a trap; interchangeability) held, and two pre-registered tests that freeze the scale of a candidate readout (the final head; the attention output) reproduce neither the trap nor the rescue; we report the transformer results as observations and make no transfer claim; in the solvable model Adam, unlike SGD, makes prompts per step matter (17× at $N = 1024$), but the transformer's dependence is steeper still and its mechanism is open.

2 Setting

Inputs $x \sim \mathcal{N}(0, I_d)$. A *skill* is a unit vector $v_p \in S^{d-1}$ with link σ; we use the normalised Hermite polynomials $\sigma_k = \mathrm{He}_k/\sqrt{k!}$, so $\mathbb{E}[\sigma_j(\langle u, x\rangle)\sigma_k(\langle w, x\rangle)] = \delta_{jk}\langle u, w\rangle^k$ for unit u, w and $k^* = k$. A *task* is a coefficient vector c with independent mean-zero entries, $\mathbb{E}c_p^2 = s_p^2$; a *prompt* consists of N pairs (x_i, y_i) with $y = \sum_p c_p \sigma(\langle v_p, x\rangle)$ and a query (x_q, y_q) from the same task. With a dictionary of P skills and single-skill prompts, skill p appears with frequency π_p.

Model A (in-context). Features $\phi_j(x) = \sigma(\langle w_j, x\rangle)$, $w_j \in S^{d-1}$, and a linear-in-label readout

$$\hat{y}_q = \sum_j \Gamma_j \, \phi_j(x_q) \, A_j, \qquad A_j = \frac{1}{N}\sum_{i=1}^{N} y_i \, \phi_j(x_i), \tag{1}$$

the MLP-plus-linear-attention architecture of Oko et al. [16]. We consider three readout protocols: *fixed* $\Gamma_j = \gamma$; *free*, a trainable scalar; and *tied*, $\Gamma_j = \|u_j\|^2$ with $w_j = u_j/\|u_j\|$ (the 2-homogeneous student of Ren et al. [17]). **Model B (in-weight)** is ordinary regression $\hat{y} = \sum_j a_j \phi_j(x)$ on the fixed task $c \equiv a$. Both are trained by online spherical SGD on the square loss with fresh data at every step: B prompts per step, step size η on the features (renormalised to the sphere after each step), step size η_Γ on a free readout initialised at Γ_0, and for the tied readout an unnormalised u with $\|u_0\|^2 = \rho_0$. We report escape times $T_{0.5}$ (steps until $|m| \geq \frac{1}{2}$) and their *flow-time* version $T_{0.5}\,\eta$, which is what the population ODEs predict; the single-feature experiments initialise at $m_0 = c\,d^{-1/2}$ exactly.

Write $g(m) = \sum_k \alpha_k^2 m^k$ for $\sigma = \sum_k \alpha_k \sigma_k$ and $V(m) = \mathbb{E}[\sigma(\langle v, x\rangle)^2 \sigma(\langle w, x\rangle)^2] = \sum_j \beta_j^2 m^j$ where $\sigma^2 = \sum_j \beta_j \sigma_j$. For σ_2: $g = m^2$, $V = 1 + 8m^2 + 6m^4$.

3 A single skill

Proposition 1 (Population loss)**.** *For $P = 1$, one feature at overlap $m = \langle w, v\rangle$, readout Γ and $\mathbb{E}c^2 = s^2$,*

$$\begin{aligned} L_A(m) &= s^2\Big[1 - 2\Gamma g(m)^2 + \Gamma^2 g(1)\Big((1 - \tfrac{1}{N})g(m)^2 + \tfrac{V(m)}{N}\Big)\Big], \\ L_B(m) &= 1 - 2a\, g(m) + a^2 g(1). \end{aligned} \tag{2}$$

Proof sketch. Conditional on the task, $A = c\,g(m) + \xi/\sqrt{N}$ with $\mathbb{E}\xi = 0$, $\mathbb{E}\xi^2 = c^2(V - g^2)$; x_q is independent of the context, and $\mathbb{E}[\sigma(\langle v, x_q\rangle)\sigma(\langle w, x_q\rangle)] = g(m)$, $\mathbb{E}[\sigma(\langle w, x_q\rangle)^2] = g(1)$. Expanding $\mathbb{E}(y_q - \hat{y}_q)^2$ gives (2). □

We verified (2) by comparing the Monte-Carlo drift $-\mathbb{E}[\partial_w L]\cdot v$ with $-\partial_m L_A$ at $m \in \{0.05, 0.1, 0.2, 0.4\}$ for $k^* \in \{1, 2, 3\}$: all 24 points agree within 2.3 s.e., and within 3% for $m \geq 0.2$ at $k^* \leq 2$ (at $k^* = 3$, $m = 0.2$ the drift is ≈ 0 and the check is noise-limited; Appendix).

The link is squared. As $N \to \infty$, $L_A = s^2[1 - 2\Gamma g^2 + \Gamma^2 g(1) g^2]$ depends on m only through $g(m)^2$, whereas L_B depends on it through $g(m)$. Near $m = 0$ the drift is $2\Gamma s^2(2 - \Gamma g(1))\, k^* \alpha_{k^*}^4 m^{2k^*-1}$ against $2ak^*\alpha_{k^*}^2 m^{k^*-1}$ for Model B: the label enters the prediction twice—through the context statistic and through the comparison with y_q—and the mean-zero task prior removes the cross terms that would provide a degree-k^* path. (A bias added to A_j would give an in-weight path for the *mean* task only, cf. Zhang et al. [19] for the linear case; we do not analyse it here.) With the escape heuristics of Ben Arous et al. [2] (drift $\eta C m^{\kappa-1}$, renormalisation drag $-\eta^2 d m/2$) the timescales are $d^{\kappa-1}$ with $\kappa = 2k^*$: for $k^* = 2$, in-weight needs $d \log d$ samples and in-context d^3.

Proposition 2 (Finite context: a drift-stable unaligned state)**.** *For σ_2 and fixed readout $\Gamma = \gamma$, $-\partial_m L_A = s^2\big[(8\gamma - 4\gamma^2(1 - \frac{1}{N}) - 24\gamma^2/N)m^3 - 16\gamma^2 m/N\big]$. For $\gamma < \gamma_1 = 2N/(N+9)$ the origin is a local minimum of L_A, the drift is repulsive for $m < m^*$ and attractive on $(m^*, 1)$, with $m^{*2} = 4\gamma/\big(2N - (N+5)\gamma\big) \approx 2\gamma/(N(1-\gamma/2))$ (for $\gamma \geq \gamma_1$ the overlap decays from every $m_0 < 1$); for general σ_k, $k \geq 2$, the same holds with $m^{*\,2(k-1)} \approx 2k\gamma/\big((2-\gamma)N\big)$ to leading order, because σ_k^2 has normalised second Hermite coefficient $\beta_2^2 = 2k^2$ (Appendix A).*

The variance $V(m)$ of the context statistic *grows* with alignment—an aligned feature has heavy tails that coincide with the teacher's—so the finite-context noise penalises alignment. A feature initialised at $m_0 \approx d^{-1/2}$ is attracted only if $d \lesssim d^* = N(8 - 4\gamma(1 - \frac{1}{N}) - 24\gamma/N)/(16\gamma)$ (≈ 31 for $\gamma = 1$, $N = 128$). For $k^* = 1$ the repulsion is a constant fraction γ/N of the attraction and is harmless. One-step analyses of pretraining do not see this term because the readout is initialised at $\gamma \propto d^{-k^*}$ [16, Lemma 18], which makes $\Gamma^2 V/N$ negligible; the phenomenon lives where Γ is order one or trained.

Remark 1 (Soft under SGD)**.** *SGD is a diffusion on the sphere; a run started below m^* escapes with a Kramers-type probability $\sim e^{-\Delta L/D_{\text{eff}}}$, $\Delta L = L(m^*) - L(m_0)$, $D_{\text{eff}} = \eta \operatorname{Var}(\partial_m\text{-gradient})/2$. At $\gamma = 1$, $N = 128$: $d = 16$ has $m_0 > m^*$ (no barrier), $d = 32$ sits on the ridge ($\Delta L/D_{\text{eff}} \approx 0.02$), $d = 64$ is trapped ($\Delta L/D_{\text{eff}} \approx 20$). Observed escapes in $3 \cdot 10^5$ steps: 3/3, 1/3, 0/3. Larger batches lower D_{eff} and* deepen *the trap.*

Proposition 3 (Readout parameterisation: three regimes—informal summary of Theorems A–B and Prop. 4, Appendix B)**.** *Consider the population gradient flow of* (2) *for $\sigma = \sigma_2$ in flow time $t = \eta \cdot$ steps, started at $m_0 = d^{-1/2}$, and let τ_{esc} be the time to reach $m = \frac{1}{2}$.*

(a) Pinned readout $\Gamma = \gamma$. *The overlap drift is $(8\gamma - 4\gamma^2(1 - \frac{1}{N}) - 24\gamma^2/N)m^3 - 16\gamma^2 m/N$ (Prop. 2). If $m_0 < m^*(\gamma, N)$ then $\tau_{\text{esc}} = \infty$; otherwise $\tau_{\text{esc}} = \Theta(m_0^{-2})$ for m_0 bounded away from m^*, diverging logarithmically as $m_0 \downarrow m^*$ (Theorem A(ii)), with an N-dependence that saturates for $N \gg \gamma d$. The condition involves N and γ only—not the number of prompts per step, which does not enter the population drift.*

(b) Free scalar readout $\dot\Gamma = -r\,\partial_\Gamma L_A$, $r = \eta_\Gamma/\eta$, $\Gamma(0) = \Gamma_0$. *The fixed point of Γ at given m is $\Gamma^*(m) = g^2/\big(g(1)[(1 - \frac{1}{N})g^2 + V/N]\big)$, approached at rate $2r\,g(1)[(1 - \frac{1}{N})g^2 + V/N] \approx 2r/N$.* Adiabatic case *($2r/N \gg$ the overlap rate): the overlap sees $L_A(m, \Gamma^*)$, whose exponent is $4k^*$ for $m^{2k^*} \ll V/N$ and $2k^*$ otherwise, and in the former regime $\tau_{\text{esc}} \propto m_0^{-(4k^*-2)}/N$ (Appendix A).* Growth-limited case *($2r/N \lesssim$ the overlap rate; E2 at $N = 16$ is adiabatic to 1%, the other cells are not): Γ lags Γ^* and τ_{esc} depends on (N, Γ_0, r) jointly; the ODE gives intermediate N-dependences (e.g. $\tau(16)/\tau(256) = 4.4$ at $d = 64$, $\Gamma_0 = 0.01$, $r = 1$, against 16 in the adiabatic $1/N$ law).*

(c) Tied readout $\Gamma = \rho = \|u\|^2$, *features $\sigma(\langle u/\|u\|, x\rangle)$. The directional velocity is $(1-m^2)\big[4gg' - \rho\,(2(1 - \frac{1}{N})gg' + V'/N)\big]$, which is exactly the pinned velocity at $\gamma = \rho$ divided by ρ (norm-independence of the attraction, 17, Lemma B.1): a tied readout behaves like a pinned readout of value $\rho(t)$ under a time change. From a small ρ_0, $\dot\rho = -4\rho\,\partial_\rho L_A$ leaves ρ near ρ_0 and $\tau_{\text{esc}} = \Theta(m_0^{-2})$ is independent of N provided $\rho_0 \le \rho^*(m_0)/4$ (Theorem A(iii)(c): $\rho(t)$ then stays within explicit factors of ρ_0 and $\big|\tau_{\text{esc}}(2k^* - 2)4k^* m_0^{2k^*-2} - 1\big| \le C_k\big(k^*\rho_0/(N m_0^{2k^*-2}) + \rho_0 + \lambda_k m_0^2\big)$, $C_2 = C_3 = 4$; the qualifier was added after E6, and the two cells where N-dependence appeared—E5's $\rho_0 = 0.1$, $N = 16$ and E6's $N = 16$, $d = 16$—are exactly those violating the hypothesis)—within it the tied readout self-selects the small-readout regime. Let $\rho^*(m) = 2Nm^2/\big(4+(N+5)m^2\big) \approx Nm^2/2$ be the value at which the directional drift vanishes. For $\rho_0 > \rho_{\text{trap}}(m_0)$ (Theorem A(iii); $\rho_{\text{trap}} \to c_2\rho^*$ with $c_2 = \frac{4}{3}$ as $m_0 \to 0$, Remark 2; at $d = 64$, $N = 16$, $\rho_{\text{trap}} = 0.178$ against $c_2\rho^* = 0.154$; between ρ^* and ρ_{trap} the overlap first falls and then escapes, proved for the small-m system), the population flow traps permanently ($m \propto t^{-2}$ while $\rho \propto t^{-1}$, so ρ never falls below the moving threshold); under SGD, noise keeps m at the noise floor while ρ decays, and the feature is eventually released—a long delay rather than a trap (E5).*

Integrating the population flow confirms the three regimes quantitatively (Appendix A): for σ_2 at $d = 512$ and $N = 32, 128, 512$ the escape times are $\infty, 745, 379$ (pinned $\gamma = 0.1$), $8.9 \cdot 10^4, 2.2 \cdot 10^4, 5.3 \cdot 10^3$ with $\tau N = 2.9, 2.8, 2.7 \times 10^6$ (free), and $39, 34, 33$ (tied); at $d = 32$ the statistic is signal-dominated for $N \gtrsim d^{k^*} = 1024$ and $\tau = 52, 37, 34, 33$ (free, $\Gamma_0 = 0.01$, $r = 1$) for $N = 32, 128, 512, 2048$. The static value Γ^* is the ridge/Wiener optimum that appears in replica analyses of trained models [7, eqs. (297)–(298)]; our point is its *dynamical* effect: the same loss yields a trap, a multiplicative N–T trade-off (that of Oko et al. [16, Remark 3]) with an exponent crossover, or no N-dependence, according to the readout. That a fast readout doubles the exponent of an in-weight problem is known [6, §5]; what is specific to the in-context problem is that the finite-context noise drives the readout towards Γ^*, so that *whether context length trades off against training time, is irrelevant, or is a hard requirement is decided by how the readout is parameterised*. One-step analyses have a readout initialised at $\gamma \propto d^{-k^*}$ [16, Lemma 18], which makes $\Gamma^2 V/N$ negligible and places them in the regime where only the product of prompts and prompt length matters.

Experiments are labelled E1–E7 in the text; they are Exps. 1b, 6, 2, 3, 7, 9 and 10 of the repository and appendix, in that order.

E1: exponents by SGD. Online SGD, σ_2, $d = 32$, $\eta = 1/d^2$, initial overlap $m_0 = c\, d^{-1/2}$, $c \in \{0.5, 0.7, 1, 1.4\}$, three seeds each; the escape time from m_0 scales as $m_0^{-(\kappa-2)}$ for $\kappa > 2$, so the slope of $\log T_{0.5}$ against $\log m_0$ estimates $\kappa_{\text{eff}} = 2 - \text{slope}$ (12 runs per cell; the estimator is coarse and the censored cells are biased low). Fig. 1 shows κ_{eff} for free ($\eta_\Gamma = \eta$ and 10η) and tied readouts at $N \in \{32, 128, 512, 4096\}$ against the population ODE of Props. 1–3 (integrated with LSODA, rtol 10^{-10}; an earlier Euler integrator overestimated times by 0.6–3.6% without changing any exponent or trap verdict) evaluated *after the run* at the same d, m_0-grid, Γ_0 and η_Γ/η (the ODE itself was fixed before the run; its evaluation at these parameters was not pre-registered): tied 4.10–4.44 (ODE 4.15–4.18); free 4.8–6.6 rising as N falls (ODE 4.85–7.0). With a censored (Tobit) fit for the three cells that contain censored runs, all twelve cells agree with the ODE within 0.5 (ordinary least squares on reached runs only: eleven within 0.6, the free $N = 32$, $\eta_\Gamma = \eta$ cell 1.0 low); the estimator is a secant over a factor 2.8 in m_0 and local slopes vary, so these are effective, not asymptotic, exponents, and the prompts per step were 32 for $N \le 128$ and 8 for $N \ge 512$ (a $B = 8$ control at $N = 128$ gives 5.38 ± 0.26 vs 5.59 ± 0.20). The ODE also reproduces absolute escape times: over the 45 fully-reached E1 cells the ratio median-SGD/ODE lies in $[0.64, 1.34]$ with 24 within 10% and 35 within 20% (no trend in N or protocol; the largest deviations are at the smallest m_0), and one censored cell (free, 10η, $N = 128$, smallest m_0) is at least 2.3× slower than predicted (Appendix C). The in-weight model at the same k^* gives a logarithmic m_0-dependence ($\kappa = 2$). κ_{eff} is a secant over a factor 2.8 in m_0 at fixed $(d, N, \Gamma_0, \eta_\Gamma/\eta)$, not an asymptotic exponent: at fixed N and $d \to \infty$ the free-readout flow gives $4k^*$ for every N with N only in the prefactor (Theorem B: $\tau_{\text{esc}} \sim e^{(4k^*-2)k^{*2}\Gamma_0^2/r}\, \mu_0^{-(4k^*-2)} d^{2k^*-1}/(4k^*(4k^*-2)N)$), while a d-free limit exists under the joint scaling $\lambda = N/d^{k^*}$, $\tilde{r} = (\eta_\Gamma/\eta)/d$, giving $\kappa(\lambda, \tilde{r}, \Gamma_0)$ whose $d \to \infty$ values (numerical; 6.45/5.41/4.96/4.83, `scripts/kappa_limits.py`) are within 0.06 of the $d = 32$ ODE numbers; its large-λ end at small Γ_0 is $2k^* + 1$ (readout growth from Γ_0), not $2k^*$. The pre-registered "$4 \to 8$" was this corner structure misstated as $d \to \infty$ limits. Two pre-registered predictions in this experiment failed and are reported as such: the learned Γ at $T_{0.5}$ is 0.18–0.25, below $\Gamma^*(0.5; N) = 0.38$–0.99 (the readout has not equilibrated at $\eta_\Gamma = \eta$; the exponent prediction does not depend on this), and a scan of $T_{0.5}$ against $d \le 48$ at $N = 128$ versus 2048 showed no N-dependence (slopes 3.49 ± 0.08 vs 3.70 ± 0.21), which is what the ODE predicts for $d \le 48$ but means the N-dependence is established only through $\kappa_{\text{eff}}(N)$ at fixed d. With the tied readout at

$\rho_0 = 1$, $N = 128$, the natural O(1)-readout version of Prop. 2, $d = 32$ escaped (3/3) while $d = 64$ was trapped (0/3, ρ decaying to 0.18).

E2: the three regimes by SGD (the prospective test of Prop. 3). To test Prop. 3 directly (Fig. 2) we fix $d = 64$, $m_0 = d^{-1/2}$, $\eta = 1/d^2$, vary $N \in \{16, 64, 256\}$ under two schemes—tokens per step fixed ($NB = 4096$) and prompts per step fixed ($B = 64$)—and run the four protocols, three seeds each, with the population escape times computed before the run. Pinned $\gamma = 1$: no escape at $N \leq 64$ in 10^6 steps (0/9, including $B = 256$), 3/6 at $N = 256$ ($d^* \approx 63$); pinned $\gamma = 0.1$: median 396k/192k/179k steps against 400k/193k/175k predicted; free: 1.78M/614k/402k against 1.82M/623k/409k, ratio $T(16)/T(256) = 4.4$ (3.6–5.0 over seed pairings) as predicted (4.44)—the growth-limited value of Prop. 3(b), not the adiabatic $1/N$ law; tied: 17.0k/15.1k/14.0k against 17.0k/16.4k/16.3k (flat within $\times 1.25$, seed pairings 0.9–1.45). Three seeds resolve medians to roughly 5%. In every protocol the two batch schemes give the same step counts, as they must at fixed step size (the population drift does not contain B); the trade-off is between context length and the number of SGD steps, and a step size scaled with B is tested in Appendix C. (The free-readout $N = 16$ cells exceeded the pre-registered 10^6-step cap, as the ODE had predicted, and were rerun to $2.5 \cdot 10^6$ steps.)

E5: controls. Varying the readout's initial value and learning rate (free: $\Gamma_0 = 0.1$ or $\eta_\Gamma = 10\eta$ at $d = 64$) changes the N-ratio from 4.4 to 9.2 and 10.7 (ODE 9.1 and 8.9; medians within 2% except the $\eta_\Gamma = 10\eta$, $N = 16$ cell, 24% slow), confirming that the free-readout exponent is a function of $(N, \Gamma_0, \eta_\Gamma/\eta)$, as Prop. 3(b) states. A pinned $\gamma = 0.01$ readout is as N-flat as the tied one (ratio 1.06, ODE 1.04). A tied readout started at $\rho_0 = 0.3 > \rho_{\text{trap}}(m_0) = 0.18$ at $N = 16$ was pre-registered to trap (the deterministic flow provably does, Theorem A(iii): $m \propto t^{-2}$, $\rho \propto t^{-1}$); under SGD all three seeds escaped after a $\approx 30\times$ delay, once ρ had decayed to 0.012—so under SGD tying converts the pinned trap into a delay, and the pre-registered wording "the repulsion survives tying" is withdrawn in favour of "is delayed by tying". Finally, scaling the step size with the batch ($\eta \propto B$, pinned $\gamma = 0.1$, $N = 64$) leaves the escape *flow time* invariant to 0.5% while steps scale as $1/\eta$: the irrelevance of prompts per step in E2 is a statement about flow time, independent of how η is chosen.

E6: a $k^* = 3$ check (Exp. 9, pre-registered P22). With $\sigma = \sigma_3$ ($g = m^3$, $V = 1 + 18m^2 + 54m^4 + 20m^6$), $\eta = 2 \cdot 10^{-4}$, $B = 64$, $d \in \{8, 16, 32\}$, three seeds: the tied readout ($\rho_0 = 0.01$, $N = 128$) gives medians 1.18/0.89/0.80 of the flow ($d = 8/16/32$; the pre-registered band was 15%, so P22a failed) and secant exponents 5.7 and 6.0 (ODE 6.5, 6.4; the $8 \to 16$ secant misses its ± 0.5 band)—an exponent near 6, against ≈ 4 at $k^* = 2$. Two post-hoc controls, declared before running: at $\eta/4$ the $d = 8$ ratio becomes 1.007 (discretisation); at $B = 256$ the $d = 32$ ratio becomes 1.14, outside the 10% band we had declared for it, so the flow is confirmed only to $\approx 15\%$ there and the $B = 64$ deficit is read as noise-assisted escape. The pinned readout $\gamma = 0.2$ has $m^* = 0.30$: $d = 8$ ($m_0 = 0.35$) escapes at $1.02\times$ the flow time and $d = 16, 32$ are censored 3/3 at 10^6 steps, as predicted. The pre-registered N-flatness of the tied readout (P22b) *failed* at $N = 16$, $d = 16$: the flow predicts $1.6\times$ the $N = 128$ time, the seeds gave 47k, 500k and censored, and at $d = 32$ all three seeds trapped (predicted: $\rho_0 = 0.01 > \rho_{\text{trap}} = 0.006$). With $k^* = 3$ the drift near m_0 is $\propto m^5$ and ρ_0 was only $2.6\times$ below ρ_{trap}, so SGD noise decides between escape and a long delay. The statement that survives is the conditional one in Prop. 3(c): N-flat for $\rho_0 \ll \rho_{\text{trap}}(m_0, N)$.

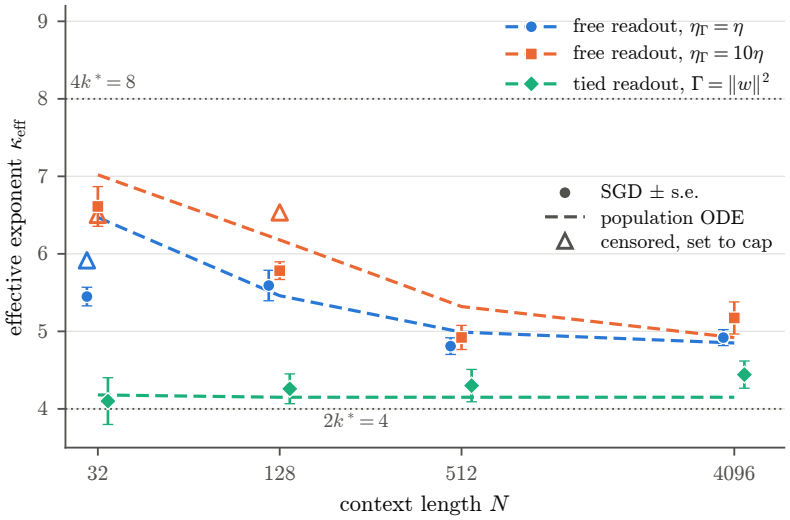


Figure 1: Effective exponent of skill acquisition for $k^* = 2$ at $d = 32$: online SGD (points, ±s.e.; triangles: censored runs set to the step cap) against the parameter-free population ODE (dashed). The tied readout stays near $2k^* = 4$ (4.1–4.4); the free readout rises towards $4k^* = 8$ as the context shortens.

E7: the regime boundaries move with d as predicted (Exp. 10, pre-registered P23). Repeating E2's protocols at $d = 128$ ($\eta = 1/d^2$, three seeds): the pinned $\gamma = 0.1$ cell at $N = 16$, which escaped at $d = 64$, is now censored 3/3 at $2.5 \cdot 10^6$ steps—the flip across $d^*(16) = 76$ of Prop. 3(a)—while $N = 64, 256$ escape at 0.99 and 1.01 of the flow time (1.78 and $1.47 \cdot 10^6$ steps; ratio 1.22, N-saturating); pinned $\gamma = 1$ is censored at every N ($d^*(256) = 63$); the tied readout stays N-flat (medians/flow $1.12, 0.97, 1.02$; $T(16)/T(256) = 1.20$, flow 1.09); the free readout at $N = 256$ takes $5.29 \cdot 10^6$ steps (0.985 of the flow) and its secant exponent across $d = 64 \to 128$ is 5.4 (flow 5.4), between the signal-dominated 4 and the adiabatic 8, and close to the large-λ, small-Γ_0 corner $2k^* + 1 = 5$ of Prop. 4. (The pre-registered secant 7.4 used the wrong $d = 64$ baseline, the $\eta_\Gamma = 10\eta$ cell; recorded as a pre-registration error.)

4 Many skills and composition

Fig. 4 summarises Exps. 2–3. With P orthogonal skills and $M \geq P$ features the loss splits, when each feature is assigned to one skill, into $\sum_p \pi_p s_p^2 \ell(m_p)$ with ℓ the single-skill loss; each overlap then follows the single-feature dynamics with rate $\propto \pi_p s_p^2 \Gamma$, so $T_p \propto 1/\pi_p$. In-weight learning of the same dictionary carries the coefficient a_p (with $a_p^2 \propto \pi_p$), hence $T_p \propto 1/a_p \propto \pi_p^{-1/2}$: in-context learning orders skills by frequency about twice as steeply—a corollary of the squared link, since the in-context drift carries $\mathbb{E}c_p^2 \pi_p$ and the in-weight drift carries a_p. **E3** ($d = 32$, $P = 16$, $\pi_p \propto p^{-1.5}$, $M = 64$, fixed $\Gamma = 0.1$, three seeds) gives slopes of $\log T_p$ versus $\log p$ of 1.44 ± 0.08 (drop midpoints; 1.53 ± 0.22 with the 0.5-crossing) for the in-context model, against 0.75 ± 0.08 for a capacity-matched in-weight baseline and 0.75–0.89 for the 2-homogeneous in-weight student of Ren et al. [17] (small initialisation, no hand-chosen readout; the lower value is limited by the logging resolution, the higher one uses a dense early log). Both in-weight baselines fit the target *collectively* at $M = 64 > d$ (every skill shared by ≥ 2 features, pooled alignment $\to 1$, no single feature above 0.9), so the in-weight slope is an empirical baseline rather than a decoupled-neuron prediction; its

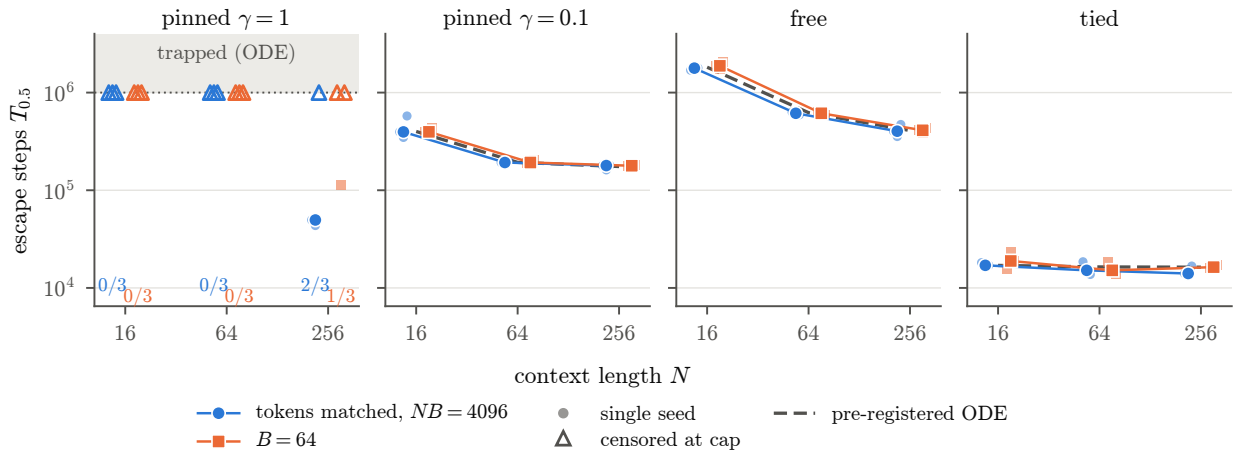


Figure 2: Escape steps $T_{0.5}$ (first step with $|m| \geq 0.5$) against context length N for the four readout protocols ($d = 64$, $\eta = 1/d^2$, $m_0 = d^{-1/2}$; shared log–log axes). Light markers are single seeds (three per cell), solid markers and lines the median over seeds, for tokens per step fixed ($NB = 4096$, circles) and prompts per step fixed ($B = 64$, squares). Open triangles are runs that had not escaped at the step cap (10^6) and are drawn at the cap; the counts under the pinned-$\gamma = 1$ panel are runs escaped out of 3. Dashed lines are the population-ODE escape times (flow time $\times d^2$), *fixed before the run*; pinned $\gamma = 1$ has no finite ODE escape time (trapped, shaded beyond the cap). The free-readout $N = 16$ cells use the reruns to $2.5 \cdot 10^6$ steps (the pre-registered cap was exceeded by the predicted 1.82M steps). At $N = 64$ both schemes have $B = 64$ and are the same three runs.

emergence times nonetheless obey $T_p \, \eta \, a_p = \text{const}$ (0.23–0.26 across skills and three learning rates), i.e. ordering by the coefficient $a_p \propto \sqrt{\pi_p}$, whereas the in-context times obey $T_p \, \eta \, \pi_p m_0^2 = \text{const}$. The constant $T_p \, \eta \, \pi_p m_0^2 = 0.73$ (IQR 0.70–0.77) across skills and seeds is close to the single-feature value (0.64, outside the IQR), i.e. the many-skill dynamics is the single-feature dynamics with rate π_p. Per-skill drops are $\approx 4\times$ sharper than in-weight (relative width 1.08 vs 4.09). The frequency-weighted alignment loss decays as $t^{-0.40 \pm 0.04}$ over the window in which skills 3–8 emerge, against $(\alpha - 1)/\alpha = 0.33$ for an idealised staircase; the window spans only a factor 3–8 in time, so the exponent depends on the emergence definition and we do not claim a scaling-law ratio.

E4: composition. Because (1) is linear in the context labels, $\hat{y}_q(c_1, c_2) = \hat{y}_q(c_1, 0) + \hat{y}_q(0, c_2)$ identically, and for independent mean-zero c_1, c_2 the loss on an additive pair is the sum of the single-skill losses. Trained on single-skill prompts only ($d = 32$, $P = 2$, $\pi = 0.75/0.25$, $M = 4$), the pair loss reaches 90% of its drop before or together with the slower skill in every run (never after)—which is forced by linearity, so this is a check of the identity under SGD, not a test; on a thresholded accuracy, where Arora and Goyal [1] predict an extra delay for tuples, the thresholded pair accuracy is approximately the product of the single-skill accuracies ($\text{acc}_{12} \approx 0.85 \, \text{acc}_1 \text{acc}_2 + 0.09$, r.m.s. 0.008 from $y = x$)—the multiplicative form observed empirically by Okawa et al. [15], and a delay of the kind Arora and Goyal [1] describe—while a Gaussian-error closed form fails (heavy tails). A product composition $\sigma_1(\langle v_1, x \rangle)\sigma_1(\langle v_2, x \rangle)$ is not learned (loss ≈ 1), as expected for a model never shown product prompts.

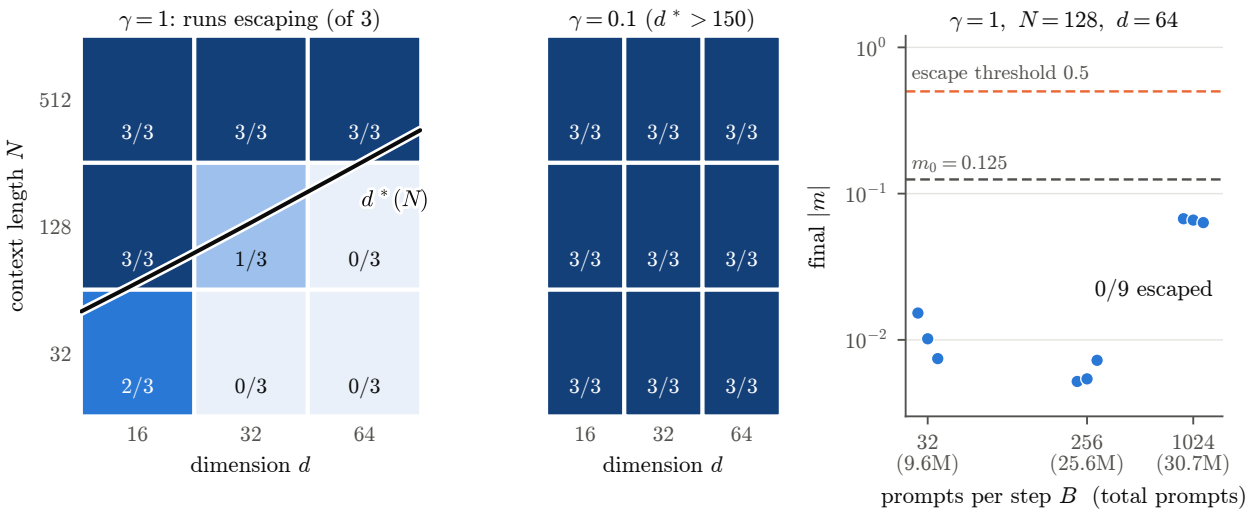


Figure 3: Left: fraction of runs escaping $m_0 = d^{-1/2}$ within $3 \cdot 10^5$ steps (fixed readout, 3 seeds per cell; curve: $d^*(N)$ of Prop. 2; for $\gamma = 0.1$ the curve lies above the grid). Right: at $\gamma = 1$, $N = 128$, $d = 64$ no run escapes for $B \in \{32, 256, 1024\}$ prompts per step (total prompts $\times 1$, $\times 2.7$, $\times 3.2$) and the final overlap falls *below* m_0: more prompts do not substitute for context length.

5 Transformers: an observation and what it excludes

We train two-layer softmax transformers (4 heads, pre-LN, Adam, online data; Fig. 5) on the same in-context task with $k^* = 2$. **Large initial alignment** ($d = 32$, MLP width 256; the best of 256 random first-layer directions already has $|\cos| = 0.43$–0.55 with v): at a fixed 8192 tokens per step the emergence step *increases* with N (1000/1400/2800 for $N = 16/64/256$)—fewer prompts per step dominate, as they must when $m_0 \gg m^*$. **Small initial alignment** ($d = 256$, width 32; $|\cos|_0 = 0.13$–0.18): $N = 16$ with $B = 64$ prompts never emerges in 30k steps (3/3); $N = 16$ with $B = 1024$ (the tokens per step of $N = 256$, $B = 64$) emerges in all three seeds (10.6–14.8k), $\approx 3\times$ later than $N = 256$ in two paired seeds (2.8–4.0k) and earlier in the third (11.6k against a run censored at 17k). Neither pre-registered extreme (a trap; interchangeability) held. These are observations, not a test of Sec. 3: the Adam step was not rescaled with B, and the readout-norm collapse on the plateau is identical in stuck and emerging runs (Fig. 5b; Appendix C).

Two null knob tests (Exps. 8 and 11; pre-registered P21, P24). If Prop. 3 acted through a scalar readout, freezing the final head, or the attention output projections of both blocks, at their O(1) initial norm should trap $N = 16$ even at $B = 1024$, and freezing them at $0.1\times$ should rescue $N = 16$, $B = 64$. Neither happened for either knob (two seeds per cell; both kill criteria met): $B = 1024$ rescued $N = 16$ under every frozen protocol that ran (5.8k/8.4k for the head, 15.0k/26.4k for the attention output), and the small scales left $N = 16$, $B = 64$ stuck. In a pre-LN transformer every block input is LayerNorm-normalised, so no scalar plays the role of the tied readout's ρ either. We make no transfer claim.

What prompts per step do. The population drift of the solvable model contains no B (Theorem A(iv)); under SGD its escape time is B-invariant to 0.5% at $N = 64$ (E5) and within 20% at $N = 1024$ (Exp. 15, where the pre-registered 15% clause P28b was missed). Under Adam the same model is strongly B-dependent where the drift is positive: at $N = 1024$ the median escape time falls

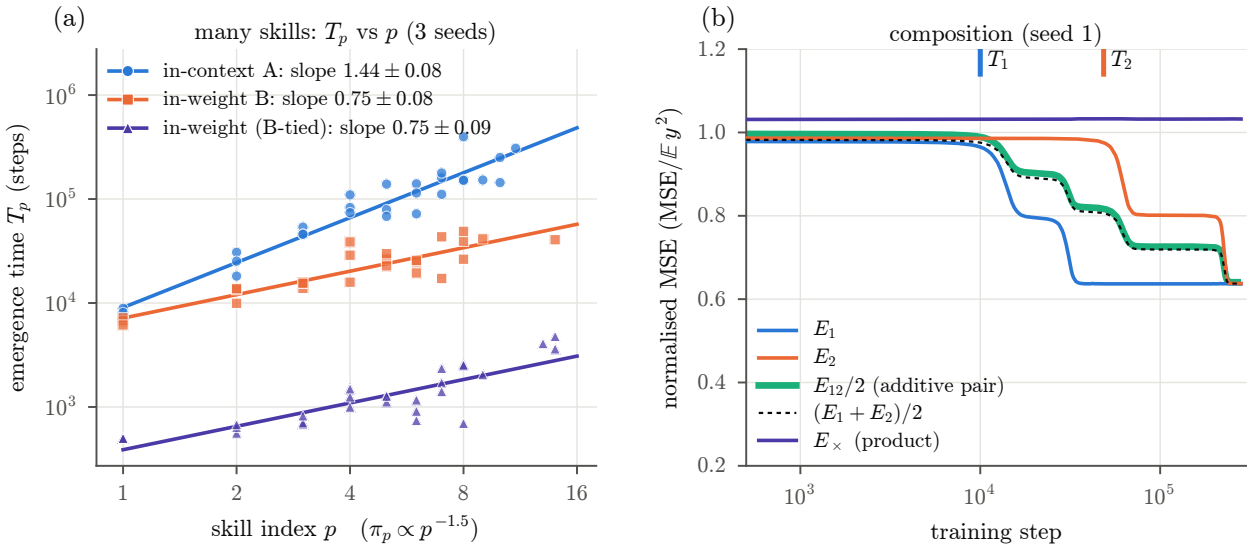


Figure 4: (a) Emergence time (drop midpoint) of skill p versus skill index for the in-context model (slope 1.44 ± 0.08), the capacity-matched in-weight baseline (slope 0.75 ± 0.08) and the 2-homogeneous in-weight student (slope 0.75–0.89), $\pi_p \propto p^{-1.5}$, three seeds each; unlearned skills omitted. (b) Composition (Exp. 3, one seed): normalised MSE on skill-1 prompts, skill-2 prompts, the additive pair (halved) and the product pair; the dotted curve is half the sum of the single-skill losses and lies on the pair curve. Plateaus at 0.64 are the fixed-readout floor $(1 - \Gamma n)^2$, not zero.

from 2537 to 146 steps between $B = 16$ and $B = 1024$ (slope -0.75 ± 0.14 of $\log T$ against $\log B$; the pre-registered $\sqrt{B}$ band $[-0.75, -0.25]$ of P28a was missed by 0.004), in line with the square-root batch-size scaling of adaptive methods [11]. The transformer's B-dependence at $N = 16$ is of a different kind: with three seeds per B (Exps. 16–17, 19; P29, P30 and both readings of P32 failed) the fraction of runs that leave the plateau within budget rises gradually—0/2 at $B = 256$ (to 80k), 1/3 at 384, 2/3 at 512 and 768, 3/3 at 1024—while the runs that do leave do so at 4–18k steps throughout, and one seed stays stuck at every $B < 1024$. Prompts per step change the probability of escaping within budget more than the time, which is neither a threshold nor a power law and is the opposite direction from the noise-driven escape of Appendix C. The mechanism of the transformer's N–B interchange is therefore open; what is excluded is that it is the population loss, the final readout, or the attention-output scale; removing the second block or all LayerNorms leaves the pattern in place (stuck at $B = 64$, rescued at $B = 1024$ in one of two seeds; Exp. 18, pre-registered P31), so neither depth nor LayerNorm is required for it. Single-feature ablations with product- and distance-based softmax statistics under SGD and Adam (Exps. 12–14) are reported in Appendix C.

6 Related work

Oko et al. [16] analyse the architecture (1) with one gradient step; the product of two correlations ($\propto m^{2Q-1}$) is in their Lemma 21 and the exponent $2Q$ in their total sample count $N_1 T_1 = \tilde{\Omega}(d^{2Q+1} r)$, framed as a consequence of correlational information; their Remark 3 states the N_1–T_1 trade-off, which Prop. 3 recovers for a shrinking readout; a readout pinned at order one lies outside their $\gamma \propto d^{-Q}$ regime, and there the trade-off is absent (Prop. 2, Fig. 3)—a complement to, not a contradiction of, their Remark 3. Nishikawa et al. [14] show that softmax's label nonlinearity lowers the *inference-time* context length to the generative exponent while one-step pretraining

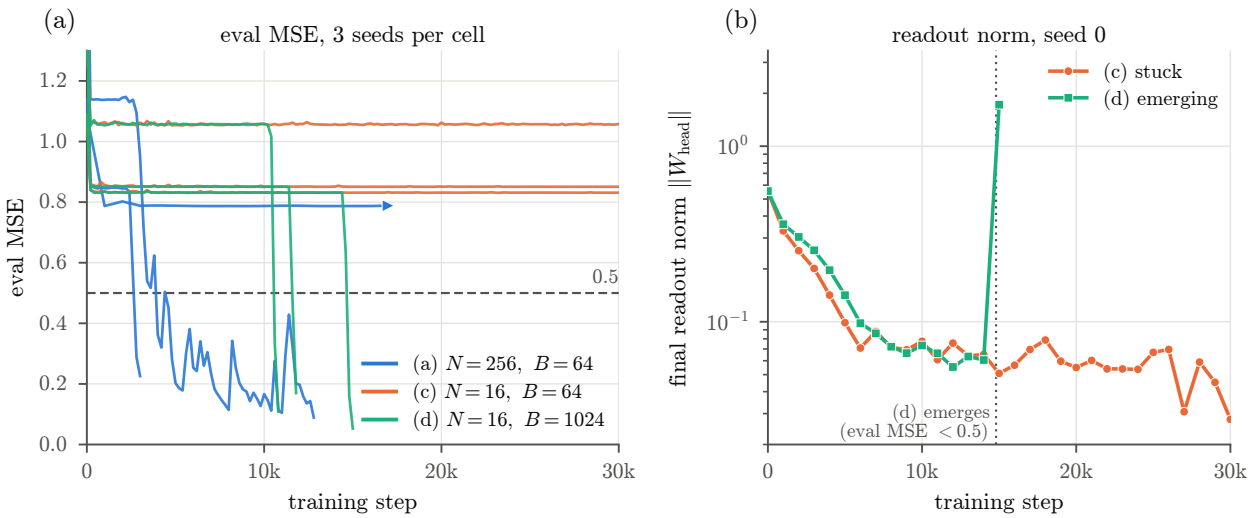


Figure 5: Two-layer softmax transformer, $d = 256$, MLP width 32, $k^* = 2$. (a) Evaluation MSE for N=256, B=64; N=16, B=64; and N=16, B=1024 (same tokens per step as the first), three seeds each; one N=256 seed was censored at 17k steps on the plateau. (b) Norm of the final readout for one stuck run (N=16, B=64) and one emerging run (N=16, B=1024, same seed): the norm collapses while the skill is unlearned; the regrowth at emergence is a single logged point (norms logged every 1000 steps).

stays at $d^{\Theta(\text{ie})}$. Gu et al. [7] derive, by the replica method for trained models with $k^* = 1$, the kernel-vs-feature context-length scalings (their feature learner has a single transition at $N = \Theta(d)$ for inference, not for feature emergence) and the static shrinkage Γ^* (their eqs. (297)–(298)); they list jointly trained readouts and $k^* \geq 2$ as open, which is where our results sit. Kim and Suzuki [9] note that at finite sample size the ICL landscape may acquire a spurious minimum at the origin (their App. C.4), a qualitative precursor of Prop. 2. Ren et al. [17] treat the additive model in-weight for even links with $k^* > 2$ with a 2-homogeneous student; our tied protocol is theirs, the norm-independence of the directional dynamics is their Lemma B.1, and $k^* = 2$ is outside their assumptions (the companion work Ben Arous et al. [3] treats the quadratic case in-weight with a student narrower than d, sample complexity $d\,\text{polylog}\,d$, and notes that the $k \geq 3$ decoupling argument does not apply at $k = 2$). The doubling of the exponent by a fast link is in Bietti et al. [6, §5] and Bietti et al. [5]; two-timescale readout dynamics in two-layer networks are analysed by Berthier et al. [4], and Ren et al. [17, Cor. 2.3] give an in-weight scaling law in which the frequency ordering is also steepened by the learning-rate schedule. Zhang et al. [19] prove, for linear regression, that the task mean is learned in-weight by the MLP and the covariance in-context. Skill-based accounts of emergence [1, 12, 13, 15] are in-weight; our composition result is the label-linear in-context counterpart. Compositions that are not output-additive need extra structure [10, 18]; He et al. [8] show that even a label-linear in-context algorithm is not free when the network must discover it against an in-weight memoriser, whereas in (1) linearity in the labels is built in.

7 Limitations

The theory is for one feature per skill (Theorems A–B hold for any $k^* \geq 2$; SGD checks are at $k^* = 2$ and 3); the SGD threshold is soft, and at $k^* = 3$ noise decides between escape and delay near

the tied trap boundary; the adiabatic $1/N$ law of Theorem B needs $N \ll d^{k^*}$ and a fast readout, so it is out of reach of online SGD at feasible sizes (at $d = 64$ the flow itself caps $\tau(16)/\tau(256)$ at 12.8 as $r \to \infty$, against 4.4 at $r = 1$ and 8.9 at $r = 10$, the values SGD reproduced; at $d = 256$ the law would need $\sim 10^9$ steps); the many-skill statements rest on decoupling, which fails for rare skills when no feature starts closest to them (2/10 seeds in Exp. 3); the Gaussian closed form for compositional accuracy is wrong; the transformer results are observational: the pre-registered N–B predictions failed in both directions, and both readout-protocol tests met their kill criteria, so no transfer claim is made; the batch-size experiments (Exps. 12–17) have two or three seeds per cell. All predictions, outcomes and deviations (including two algebra slips corrected after pre-registration, one wrong pre-registered baseline, and two ill-posed estimators) are listed verbatim in the appendix.

Reproducibility and disclosure. Code, pre-registration document and all runs: [repo]. AI assistance was used for literature search, implementation, derivation checking and drafting; every derivation was verified against the Monte-Carlo drift, and every citation against the full text.

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A Derivations

Hermite algebra. For unit u, w with $\langle u, w\rangle = m$ and $z_u = \langle u, x\rangle$, $z_w = \langle w, x\rangle$ jointly Gaussian with correlation m, $\mathbb{E}[\sigma_j(z_u)\sigma_k(z_w)] = \delta_{jk} m^k$. For $\sigma = \sigma_2 = (z^2-1)/\sqrt{2}$: $\sigma_2^2 = \sqrt{6}\,\sigma_4 + 2\sqrt{2}\,\sigma_2 + 1$, hence $V(m) = \mathbb{E}[\sigma_2(z_u)^2\sigma_2(z_w)^2] = 6m^4+8m^2+1$ and $g(m) = m^2$. For $\sigma_1 = z$: $g = m$, $V = 1+2m^2$.

Population loss (Prop. 1). With $A = \frac{1}{N}\sum_i y_i\sigma(\langle w, x_i\rangle)$ and $y_i = c\,\sigma(\langle v, x_i\rangle)$, $\mathbb{E}[A \mid c] = c\,g(m)$ and $\mathrm{Var}[A \mid c] = c^2(V(m)-g(m)^2)/N$. Since x_q is independent of the context, $\mathbb{E}[y_q\hat{y}_q] = \Gamma\,\mathbb{E}c^2\,g(m)^2$ and $\mathbb{E}[\hat{y}_q^2] = \Gamma^2 g(1)\,\mathbb{E}c^2\left(g^2 + (V-g^2)/N\right)$, giving (2).

Drift (Prop. 2). For σ_2, $\Gamma=\gamma$, $s^2=1$: $L_A = 1-2\gamma m^4+\gamma^2\big[(1-\frac{1}{N})m^4+(1+8m^2+6m^4)/N\big]$, so $-\partial_m L_A = \big(8\gamma-4\gamma^2(1-\frac{1}{N})-24\gamma^2/N\big)m^3-16\gamma^2 m/N$, which is negative for $m<m^*$ with $m^{*2}=16\gamma/\big(N(8-4\gamma(1-\frac{1}{N})-24\gamma/N)\big)$. (An earlier draft of this derivation dropped the $24\gamma^2/N$ term and had a factor 2 in the cubic coefficient; the closed-form loss, which the Monte-Carlo drift check verifies directly, was unaffected.)

General σ_k. The normalised second Hermite coefficient of σ_k^2 is $\beta_2=\sqrt{2}\,k$ (checked by quadrature for $k\le 4$), so the finite-context repulsion is $-\Gamma^2\beta_2^2\,2m/N=-4k^2\Gamma^2 m/N$ to leading order while the attraction is $2\Gamma(2-\Gamma g(1))\,k\,m^{2k-1}$: the origin is drift-stable below $m^{*\,2(k-1)}\approx 2k\Gamma/((2-\Gamma)N)$ for every $k\ge 2$ (for $k=1$ the two terms have the same power and the repulsion is a constant fraction). At $k=3$, $\Gamma=0.1$, $N=128$ the trap sets in at $d^*\approx 17$ (against ≈ 607 at $k=2$), consistent with the negative Monte-Carlo drift observed for $k^*=3$ at $d=16$. The step size $\eta\propto d^{-2}$ is the critical scaling only for $k^*=2$.

Tied = pinned up to a time change. The tied directional velocity equals the pinned velocity at $\gamma=\rho$ divided by ρ; with $\dot\rho=-4\rho\,\partial_\rho L_A$ the pair (m,ρ) above threshold obeys $m\propto t^{-2}$, $\rho\propto t^{-1}$ (LSODA to $t=10^6$), and the moving threshold $\rho^*(m)\approx Nm^2/2\propto t^{-4}$ falls faster than ρ: the population flow traps permanently.

Escape-time scalings in Prop. 3. (a) With $\Gamma=\gamma$ fixed and $m_0>m^*$ the drift near the origin is $\approx c_\gamma m^{2k^*-1}$ with $c_\gamma=\gamma(8-4\gamma)k^*\alpha_{k^*}^4/2$ for $k^*=2$, so $\tau_{\rm esc}\asymp m_0^{-(2k^*-2)}/c_\gamma$ up to the N-dependent correction $-16\gamma^2 m/N$, which saturates as $N\to\infty$. (b) Along Γ^*, for $g^2\ll V/N$, $L_A(m,\Gamma^*)\approx s^2[1-Ng(m)^4/(g(1)V(m))]$ and $V(m)\to\beta_0^2=g(1)^2$ as $m\to 0$, so the drift is $\approx s^2N\,4k^*\alpha_{k^*}^8 m^{4k^*-1}/g(1)^3$: $\tau_{\rm esc}\asymp m_0^{-(4k^*-2)}/N$, i.e. $\tau_{\rm esc}\propto d^{2k^*-1}/N$—the multiplicative N–τ trade-off. The condition for this regime at the initial overlap is $m_0^{2k^*}\ll V/N$, i.e. $d^{k^*}\gg N$; for $d^{k^*}\ll N$ the signal-dominated form $1-g^2/g(1)$ applies and $\tau_{\rm esc}\asymp m_0^{-(2k^*-2)}$ is N-independent. (c) For the tied readout the attraction term $4gg'$ carries no ρ, so $\tau_{\rm esc}\asymp m_0^{-(2k^*-2)}/(4k^*\alpha_{k^*}^4)$ independently of N as long as $\rho V'/N\ll 4gg'$ along the trajectory, which holds from $\rho_0\le\rho^*(m_0)/4$ (Theorem A(iii)(c), with the N-dependence bounded by $C_k k^*\rho_0/(Nm_0^{2k^*-2})$), which for $k^*=2$ reads $\rho_0\ll Nm_0^2/2$; for $k^*=3$, $\rho_{\rm trap}\propto m_0^4$ and $\rho_0=0.01$ already violates it at $N=16$, $d\ge 16$ (Exp. 9). At $d=512$, $k^*=2$ the three regimes give $\tau\propto\{\infty$ or N-saturating, N^{-1}, $N^0\}$, matching the integrated flows ($\infty/745/379$; $8.9\cdot10^4/2.2\cdot10^4/5.3\cdot10^3$; $39/34/33$ for $N=32/128/512$).

Population ODEs (Prop. 3). Spherical gradient flow for the overlap and plain flow for the readout: free readout $\dot m=-(1-m^2)\partial_m L_A$, $\dot\Gamma=-r\,\partial_\Gamma L_A$ with $r=\eta_\Gamma/\eta$; tied readout $\dot m=(1-m^2)\big[4gg'-\rho(2(1-\frac{1}{N})gg'+V'/N)\big]$, $\dot\rho=-4\rho\,\partial_\rho L_A$. The chain rule through $\hat u=u/\|u\|$ gives the directional velocity a factor $1/\|u\|^2$ that cancels the $\rho=\|u\|^2$ carried by the attraction term but not the ρ carried by the repulsion. Escape times $T_{0.5}$ to $m=0.5$ from $m_0=c\,d^{-1/2}$ were integrated with an adaptive step (1% relative change per step) and the exponent read off as $\kappa_{\rm eff}=2-\partial\log T/\partial\log m_0$ over $c\in\{0.5,0.7,1,1.4\}$, exactly as for the SGD data.

Kramers estimate. $D_{\rm eff}=\eta\,{\rm sd}^2/2$ with the per-step standard deviation of the m-component of the batch gradient measured by Monte Carlo (0.33–0.37 at $B=32$); the estimate ignores the spherical correction term, which is of the same order as the quoted $d=32$ ratio, so only the $d=64$ value (≈ 20) is meaningful.

Monte-Carlo drift check. $d = 16$, $N = 128$, $\gamma = 0.1$, $a = 1$, $5 \cdot 10^4$ batches of 64 prompts per point. Ratios MC/formula at $m = 0.2$ and 0.4: Model B $k^* = 1, 2, 3$: 0.998–1.00; Model A $k^* = 1$: 1.000; $k^* = 2$: 0.972, 0.998; $k^* = 3$: 0.88 ± 0.76 (drift ≈ 0), 1.02. Model A $k^* = 3$ at $\gamma = 0.1$, $N = 128$ has negative drift for $m \le 0.2$.

B Theorems and proofs

All closed forms in this section are checked numerically by `scripts/verify_theorems.py` (61 checks: thresholds, the exact escape time against LSODA, the time-change identity, the t^{-2}/t^{-1} asymptotics, Theorem B at d up to $6.5 \cdot 10^4$, the corner exponents).

Standing notation. $\sigma = \sigma_k$, $k \ge 2$, so $\alpha_k = 1$, $g(m) = m^k$, $g(1) = 1$. Write $\sigma_k^2 = \sum_{i=0}^{k} \beta_{2i}\sigma_{2i}$; the linearisation formula $\mathrm{He}_k^2 = \sum_j j!\binom{k}{j}^2 \mathrm{He}_{2k-2j}$ gives $\beta_{2i}^2 = \big((k-i)!\binom{k}{i}^2\big)^2(2i)!/k!^2 \ge 0$, in particular $\beta_0^2 = 1$, $\beta_2^2 = 2k^2$, $\beta_{2k}^2 = \binom{2k}{k}$, and $V(m) = \sum_{i=0}^{k} \beta_{2i}^2 m^{2i}$, $V'(1) = 4k\,\mathbb{E}[\sigma_k^2\sigma_{k-1}^2]$ (= 40 for $k = 2$). Put $G = g^2 = m^{2k}$, $D(m) = (1 - \frac{1}{N})m^{2k} + V(m)/N$, so that

$$L(m,\Gamma) = s^2\big[1 - 2\Gamma G + \Gamma^2 D\big], \quad \partial_m L = s^2(\Gamma^2 D' - 2\Gamma G'), \quad \partial_\Gamma L = 2s^2(\Gamma D - G), \quad \Gamma^*(m) = G/D.$$

Two elementary facts are used throughout: $D \ge V(0)/N = 1/N$, and $mV' \le 2kV$ (hence $mD' \le 2kD$), because V has degree $2k$ and nonnegative coefficients. Flows: pinned $\dot m = -(1 - m^2)\partial_m L(m,\gamma)$; free $\dot m = -(1-m^2)\partial_m L$, $\dot\Gamma = -r\partial_\Gamma L$; tied $\dot m = -(1-m^2)\partial_m L(m,\rho)/\rho$, $\dot\rho = -4\rho\,\partial_\rho L(m,\rho)$. The interval $(0,1)$ is invariant for m in all three ($g'(0) = V'(0) = 0$ for $k \ge 2$), $\tau_{\mathrm{esc}} = \inf\{t : m(t) \ge \frac{1}{2}\}$, and t is flow time (η·steps).

Theorem A (Finite-context trap; population flow, $\sigma = \sigma_k$, $k \ge 2$)**.** (i) Pinned readout $\Gamma \equiv \gamma > 0$. $\dot m = s^2\gamma(1-m^2)\,m\,Q_\gamma(m^2)$ *with*

$$Q_\gamma(x) = 2k\big(2 - \gamma(1 - \tfrac{1}{N})\big)x^{k-1} - \frac{\gamma}{N}\sum_{i=1}^{k} 2i\beta_{2i}^2 x^{i-1}.$$

Let $\gamma_1 = 2N/\big(N - 1 + V'(1)/2k\big)$. *If* $\gamma < \gamma_1$, Q_γ *has exactly one positive root* x^*, *and* $x^* \in (0,1)$; *with* $m^* = \sqrt{x^*}$, $m(t) \downarrow 0$ *(exponentially, at rate* $\to 4k^2s^2\gamma^2/N$*) if* $m_0 < m^*$, *and* $m(t) \uparrow 1$ *with* $\tau_{\mathrm{esc}} < \infty$ *if* $m^* < m_0 < \frac{1}{2}$. *If* $\gamma \ge \gamma_1$, $m(t) \to 0$ *from every* $m_0 \in (0,1)$. *As* $N \to \infty$ *at fixed* $\gamma \in (0,2)$, $m^{*\,2(k-1)} = \frac{2k\gamma}{(2-\gamma)N}\big(1 + O(N^{-1/(k-1)})\big)$. *For* $k = 2$: $\gamma_1 = \frac{2N}{N+9}$ *and* $m^{*2} = \frac{4\gamma}{2N-(N+5)\gamma}$.

(ii) Exact escape time, $k = 2$. *Let* $C = \gamma(8 - 4\gamma - 20\gamma/N)$, $B = 16\gamma^2/N$ ($C > B \iff \gamma < \gamma_1$; *the constants in the* $O(\cdot)$ *terms below are uniform for* $\gamma \le (1-\delta)\gamma_1$, $\delta > 0$) *and* $m^{*2} < m_0^2 < \frac{1}{4}$. *Then*

$$\tau_{\mathrm{esc}} = \frac{F(\frac{1}{4}) - F(m_0^2)}{2s^2}, \qquad F(u) = -\frac{\ln u}{B} - \frac{\ln(1-u)}{C-B} + \frac{C\ln(Cu - B)}{B(C-B)}.$$

Hence $2s^2Cm_0^2\,\tau_{\mathrm{esc}} = 1 + O\big(m_0^2\ln\frac{1}{m_0} + m^{*2}/m_0^2\big)$, *and* $\tau_{\mathrm{esc}} \sim \frac{C}{2s^2B(C-B)}\ln\frac{1}{m_0^2 - m^{*2}}$ *as* $m_0 \downarrow m^*$.

(iii) Tied readout. *(a)* $\dot m_{\mathrm{tied}}(m,\rho) = \dot m_{\mathrm{pinned}}(m;\gamma{=}\rho)/\rho$ *identically. Hence, if* $(m,\rho)(t')$ *is the tied solution and* $t = \int_0^{t'} du/\rho(u)$, *then* $m(t)$ *solves the pinned flow with the time-dependent value* $\gamma(t) = \rho$ *(i.e.* $dt' = \rho\,dt$*); for* ρ *frozen at* ρ_0, $\tau_{\mathrm{tied}} = \rho_0\,\tau_{\mathrm{pinned}}(\gamma{=}\rho_0)$. *(b) Let* $m_0 < \frac{1}{2}$, $D_0 = D(m_0)$ *and* $a = k^2(k-1)(1-m_0^2)/(ND_0)$. *If* $a > 1$ *and*

$$\rho_0 \;>\; \rho_{\mathrm{trap}}(m_0) := \frac{k(k-1)\,m_0^{2k-2}}{D_0\,(a-1)}, \tag{3}$$

the tied flow is trapped for ever: $m(t) \le m_0$ for all t, $m(t) \to 0$, and, as $t \to \infty$, $\rho(t)\,t \to N/(8s^2)$ and $m(t)\,t^{k^2/2} \to c \in (0,\infty)$; for $k=2$, $m \propto t^{-2}$, $\rho \propto t^{-1}$. Since $\rho^*(m) := 2G'/D' \asymp Nm^{2k-2}/k$, $\rho/\rho^*(m) \to \infty$: the threshold recedes faster than ρ decays. (c) (Tied readout: N-flat escape for small ρ_0.) *Let $m_0 \in (0,\frac12)$, $\rho_0 \le \frac14\rho^*(m_0)$, and put $\varepsilon = k\rho_0/(Nm_0^{2k-2})$, $X = s^2(2k-2)\,4k\,m_0^{2k-2}$, $\bar v_k = V'(\frac12)/(2k^2)$, $c_1 = 2V(\frac12)/(k(k-1))$. Then m increases to $\frac12$ in finite time $\tau_{\rm tied}$, $\rho/\rho^*(m) < \frac13$ throughout, and for $t \le \tau_{\rm tied}$*

$$\frac{\rho_0}{1+c_1\rho_0/(Nm_0^{2k-2})} \;\le\; \rho(t) \;\le\; \rho_0\Big(\frac{1-m_0^2}{1-m(t)^2}\Big)^{3/(2k)} \;\le\; (\tfrac43)^{3/(2k)}\rho_0. \tag{4}$$

With $\rho_{\min},\rho_{\max}$ the extremes of ρ on $[0,\tau_{\rm tied}]$ and $\ell_k = (2k-2)m_0^{2k-4}\int_{m_0}^{1/2} m^{3-2k}dm$ ($= 2\ln\frac{1}{2m_0}$ for $k=2$, $\le \frac{k-1}{k-2}$ for $k\ge 3$),

$$\begin{aligned} X\tau_{\rm tied} \;&\ge\; \big(1-(2m_0)^{2k-2}\big)\Big(1+\tfrac{(1-\frac1N)\rho_{\min}}{2}\Big) + \big(1-(2m_0)^{4k-4}\big)\frac{k\rho_{\min}}{2Nm_0^{2k-2}},\\ X\tau_{\rm tied} \;&\le\; 1+\tfrac43\ell_k m_0^2 + \rho_{\max} + \bar v_k\frac{k\rho_{\max}}{Nm_0^{2k-2}}, \end{aligned} \tag{5}$$

hence $|X\tau_{\rm tied}-1| \le C_k\big(\varepsilon+\rho_0+\lambda_k m_0^2\big)$ with $C_k = \max\{4, (\frac43)^{3/(2k)}\bar v_k\}$, $\lambda_2 = 1+\ln\frac{1}{2m_0}$, $\lambda_k = 1$ ($k\ge 3$). For $k=2$: $c_1 = \frac{27}{8}$, $\bar v_2 = \frac{11}{8}$, $C_2 = 4$; for $k=3$: $c_1 = \frac{49}{16}$, $\bar v_3 = \frac{65}{24}$, $C_3 = 4$. The context length enters (4)–(5) only through ε and $1-\frac1N$, so for $\varepsilon \to 0$ the escape time is N-flat.

(iv) *For every number B of prompts per step the expected minibatch gradient is ∇L; the population flows above, and hence $m^*(\gamma,N,k)$, γ_1 and (ii)–(iii), do not depend on B.*

Lemma A.1 (Monotone signal-to-noise)**.** *$R(m) = G^2/D$ is strictly increasing on $(0,1]$, and $\rho^*(m) = 2G'/D'$ satisfies $\rho^* > \Gamma^*$ and is strictly increasing.*

Proof. $mR'/R = 4k - mD'/D \ge 2k > 0$. $\rho^* > \Gamma^* \iff 2G'D > GD' \iff 4kD > mD'$. Finally $\rho^* = 4k/\big(2k(1-\frac1N) + N^{-1}\sum_i 2i\beta_{2i}^2 m^{2i-2k}\big)$ and each m^{2i-2k}, $i<k$, is strictly decreasing. □

Proof of Theorem A. (i) Insert $G' = 2km^{2k-1}$ and $V'(m)/m = \sum_{i\ge1} 2i\beta_{2i}^2 m^{2i-2}$ in $-\partial_m L$. The coefficients of Q_γ of degree $< k-1$ are strictly negative ($\beta_{2i}^2 > 0$), and $Q_\gamma(0) = -4k^2\gamma/N < 0$. By Descartes' rule Q_γ has at most one positive root; it has one iff the leading coefficient $2k[2-\gamma(N-1+\beta_{2k}^2)/N]$ is positive, and the root lies in $(0,1)$ iff $Q_\gamma(1) = 2k[2-\gamma(N-1+V'(1)/2k)/N] > 0$, i.e. $\gamma < \gamma_1$ (this implies the former, as $V'(1)/2k > \beta_{2k}^2$). Then $Q_\gamma < 0$ on $(0,x^*)$ and > 0 on (x^*,∞); if $\gamma \ge \gamma_1$, $Q_\gamma < 0$ on $(0,1)$. The one-dimensional phase portrait follows: on $[m_0,\frac12] \subset (m^*,1)$, $\dot m \ge c > 0$; on $(0,m_0] \subset (0,m^*)$, $\dot m/m \le -c' < 0$ and $\dot m/m \to s^2\gamma Q_\gamma(0)$. For the asymptotics, the root equation $2k(2-\gamma(1-\frac1N))\,x^{*k-1} = \frac{\gamma}{N}\sum_i 2i\beta_{2i}^2 x^{*i-1}$ forces $x^* \to 0$ and gives $x^{*k-1} = \frac{4k^2\gamma}{2k(2-\gamma)N}(1+O(x^*+N^{-1}))$ with $x^* = O(N^{-1/(k-1)})$. For $k=2$, $Q_\gamma(x) = (Cx-B)/\gamma$.

(ii) With $u = m^2$, $\dot u = 2s^2u(1-u)(Cu-B)$, and $\frac{1}{u(1-u)(Cu-B)} = -\frac{1}{Bu} + \frac{1}{(C-B)(1-u)} + \frac{C^2}{B(C-B)(Cu-B)}$; integrate from u_0 to $\frac14$. Expanding F in $\varepsilon = B/(Cu_0)$ gives $2s^2(F(\frac14)-F(u_0)) = \frac{1}{Cu_0}(1+O(\varepsilon)) + O(C^{-1}\ln\frac{1}{u_0})$; the logarithmic singularity at $u_0 = B/C$ is the last term of F.

(iii)(a) $\partial_m L(m,\gamma) = s^2\gamma(\gamma D' - 2G')$, so $\dot m_{\rm pinned}/\gamma = s^2(1-m^2)(2G'-\gamma D')$, which is the tied velocity $-(1-m^2)\partial_m L(m,\rho)/\rho$ at $\gamma=\rho$ (the $1/\rho$ is the chain rule through $w = u/\|u\|$, $\rho = \|u\|^2$; cf. 17, Lemma B.1). Reparametrising time by $dt = dt'/\rho$ gives the claim.

(b) Write $z = m^{-(2k-2)}$. While $m \le \bar m$ ($\bar m \downarrow m_0$ at the end), using $1-m^2 \le 1$ in the attraction, $1-m^2 \ge 1-\bar m^2$ and $V'(m)/m \ge 4k^2$ in the repulsion, dropping the term $-2k(1-\frac1N)\rho m^{2k-2}$, and

using that D is increasing,

$$\frac{d\ln m}{dt} \le s^2\Big[4km^{2k-2} - (1-\bar m^2)\frac{4k^2\rho}{N}\Big], \qquad \dot\rho = 8s^2\rho(G-\rho D) \ge -8s^2 D(\bar m)\rho^2,$$

so $\rho \ge \rho_0/E$ with $E(t) = 1 + 8s^2 D(\bar m)\rho_0 t$, and $\dot z \ge -(2k-2)4ks^2 + (2k-2)(1-\bar m^2)\frac{4k^2s^2}{N}\rho\, z$. Comparison with the linear equation gives $z(t) \ge (z_0 - K)E^a + KE$ with $K = k(k-1)/(\rho_0 D(\bar m)(a-1))$ (and a evaluated at $\bar m$). By (3) $z_0 > K$ for $\bar m$ close to m_0, so the right side is increasing in $E \ge 1$ and $z(t) \ge z_0$: $m \le m_0 < \bar m$ for all t (continuity argument), $m \to 0$, and $\rho z \ge \rho_0(z_0 - K)E^{a-1} \to \infty$. Restarting the argument at a late time t_0 (where $ND(m(t_0)) \to 1$, so (3) holds with a arbitrarily close to $k^2(k-1)$) gives $m^{2k-2} \le Ct^{-a'}$ for every $a' < k^2(k-1)$. Hence m^2 and G/ρ are integrable in t, so $\frac{d}{dt}\frac{1}{\rho} = 8s^2(D - G/\rho) = \frac{8s^2}{N} + (\text{integrable})$ gives $\frac{1}{\rho} = \frac{8s^2 t}{N} + c_1 + o(1)$. Then $\frac{d\ln m}{dt} = -\frac{4k^2s^2}{N}\rho(1 + O(m^2)) + O(m^{2k-2}) = -\frac{k^2}{2t} + (\text{integrable})$, so $m\, t^{k^2/2}$ converges to a positive limit.

(c) Write $q = \rho/\rho^*(m)$. Since $\partial_m L(m,\rho) = s^2\rho(\rho D' - 2G')$ and $\rho^* = 2G'/D'$, the tied flow is

$$\dot m = s^2(1-m^2)\,2G'(1-q) = 4ks^2(1-m^2)m^{2k-1}(1-q), \qquad \dot\rho = 8s^2\rho\,(G - \rho D),$$

and $1/\rho^* = \frac12(1 - \frac1N) + V'(m)/(4kNm^{2k-1})$; the first summand is the $O(\rho)$ term $2k(1-\frac1N)m^{2k-1}\rho$ of $\rho D'$, the second the noise term. As $V'(m)/m = \sum_{i\ge1} 2i\beta_{2i}^2 m^{2i-2}$ is increasing from $4k^2$, for $m \le \frac12$

$$\rho\Big[\tfrac12(1-\tfrac1N) + \frac{k}{Nm^{2k-2}}\Big] \le q \le \rho\Big[\tfrac12(1-\tfrac1N) + \bar v_k\frac{k}{Nm^{2k-2}}\Big], \qquad \text{and} \quad \rho^* < \frac{2}{1-1/N}. \tag{6}$$

Bootstrap. Let $T = \sup\{t \le \tau_{\text{tied}} : q < \frac13 \text{ on } [0,t]\}$ (with $\tau_{\text{tied}} = \infty$ allowed a priori); $T > 0$ as $q(0) \le \frac14$. On $[0,T)$, $\dot m > 0$, so $m \ge m_0$, and $\dot m \ge 2ks^2 m_0^{2k-1} > 0$ while $m \le \frac12$, so $\tau_{\text{tied}} < \infty$ if $T = \tau_{\text{tied}}$. With m as the variable on $[0,T)$,

$$\frac{d\ln\rho}{dm} = \frac{\dot\rho}{\rho\dot m} = \frac{2(G-\rho D)}{k(1-m^2)m^{2k-1}(1-q)} \le \frac{2m}{k(1-m^2)(1-q)} \le \frac{3m}{k(1-m^2)},$$

and integrating from m_0 gives the upper bound in (4); with $m \le \frac12$ and $k \ge 2$, $\rho \le (\frac43)^{3/4}\rho_0 < 1.241\rho_0$. By Lemma A.1 $\rho^*(m) \ge \rho^*(m_0)$, so $q \le 1.241\rho_0/\rho^*(m_0) \le 0.32 < \frac13$ on $[0,T)$; by continuity $T = \tau_{\text{tied}}$, which proves the first two claims and the upper bound. *Lower bound.* Similarly

$$\frac{d}{dm}\frac{1}{\rho} = \frac{2(D - G/\rho)}{k(1-m^2)m^{2k-1}(1-q)}, \qquad D - \frac{G}{\rho} = \frac{V}{N} - m^{2k}\Big(\frac{1}{\rho} - 1 + \frac1N\Big) \le \frac{V}{N},$$

because $\rho \le 0.32\rho^*(m_0) < 0.64/(1-\frac1N)$ by (6). With $1/(1-q) \le \frac32$ (from $q < \frac13$) and $V/(1-m^2) \le \frac43 V(\frac12)$ (increasing in m), $\frac{d}{dm}\frac{1}{\rho} \le \frac{4V(1/2)}{kNm^{2k-1}}$, and integrating from m_0, $\frac{1}{\rho} \le \frac{1}{\rho_0} + \frac{4V(1/2)}{k(2k-2)Nm_0^{2k-2}} = \frac{1}{\rho_0} + \frac{c_1}{Nm_0^{2k-2}}$. (The ratio ρ/ρ_0 is not $1 + O(\rho_0)$: as $\rho_0 \to 0$, $d\ln\rho/dm \to 2m/(k(1-m^2))$, so $\rho(\tau_{\text{tied}})/\rho_0 \to (\frac43(1-m_0^2))^{1/k}$; the exponent in (4) is $1/(k(1-\sup q))$ with $\sup q \le \frac13$. This growth affects τ_{tied} only through q.) *Escape time.* Since m is increasing, $\tau_{\text{tied}} = \int_{m_0}^{1/2} dm/\dot m$, i.e.

$$X\tau_{\text{tied}} = (2k-2)m_0^{2k-2}\int_{m_0}^{1/2}\frac{m^{1-2k}\,dm}{(1-m^2)(1-q)}, \qquad (2k-2)m_0^{2k-2}\int_{m_0}^{1/2} m^{1-2k}dm = 1 - (2m_0)^{2k-2}.$$

Lower bound: use $\frac{1}{(1-m^2)(1-q)} \ge 1 + q$, the left inequality of (6) with $\rho \ge \rho_{\min}$, and the identity $(2k-2)m_0^{2k-2}\int_{m_0}^{1/2} m^{3-4k}dm = \big(1 - (2m_0)^{4k-4}\big)/(2m_0^{2k-2})$. Upper bound: for $m \le \frac12$, $q \le \frac13$,

$\frac{1}{1-m^2} \le 1+\frac{4}{3}m^2$ and $\frac{1}{1-q} \le 1+\frac{3}{2}q$, so

$$\frac{1}{(1-m^2)(1-q)} \le 1+\tfrac{4}{3}m^2+\tfrac{3}{2}q\big(1+\tfrac{4}{3}m^2\big) \le 1+\tfrac{4}{3}m^2+2q;$$

the m^2 term integrates to $\frac{4}{3}\ell_k m_0^2$, and the right inequality of (6) with $\rho \le \rho_{\max}$ gives $2q \le \rho_{\max} + 2\bar v_k \rho_{\max} k/(N m^{2k-2})$, whose two terms integrate to at most $\rho_{\max}$ and $\bar v_k \rho_{\max} k/(N m_0^{2k-2})$. This is (5). Finally, the left side of (5) is $\ge 1-(2m_0)^{2k-2} \ge 1-4m_0^2 \ge 1-4\lambda_k m_0^2$, and by (4) the right side is $\le 1+\frac{8}{3}\lambda_k m_0^2 + 1.25\rho_0 + (\frac{4}{3})^{3/(2k)}\bar v_k \varepsilon$ (using $\frac{4}{3}\ell_2 \le \frac{8}{3}\lambda_2$, and $\frac{4}{3}\ell_k \le \frac{4}{3}\cdot\frac{k-1}{k-2} \le \frac{8}{3}$ for $k \ge 3$), which gives the bound on $|X\tau_{\text{tied}} - 1|$.

(iv) The per-prompt losses in a step are i.i.d. with mean L; the gradient of their average has mean ∇L for every B (differentiation under $\mathbb{E}$ is justified by polynomial moments), and B enters only the covariance, which is $\propto 1/B$. □

Remark 2 (The trap boundary is $c_k\rho^*$, not ρ^*)**.** *Condition* (3) *is sharp as $m_0 \to 0$: since $ND_0 \to 1$ and $a \to k^2(k-1)$, it reads $\rho_0 > c_k\, N m_0^{2k-2}/k\,(1+o(1))$ with $c_k = 1 + 1/(k^3-k^2-1)$ ($c_2 = \frac{4}{3}$, $c_3 = \frac{18}{17}$), and in the small-m system $\dot m = s^2 m(4km^{2k-2} - 4k^2\rho/N)$, $\dot\rho = -8s^2\rho^2/N$ (exactly $z = (z_0-K)E^a + KE$ with $a = k^2(k-1)$, $ND_0 = 1$) z reaches 0 in finite time, i.e. the flow escapes, for every $\rho_0 < c_k N m_0^{2k-2}/k$. So for $\rho^*(m_0) < \rho_0 < c_k\rho^*(m_0)$ the overlap first* decreases *and then escapes: "$\rho_0 > \rho^*(m_0) \Rightarrow$ trap" is false. LSODA bisection of the exact flow gives the boundary $\rho_0/\rho^*(m_0) =$ 1.3333 (k=2, m_0=10^{-3}), 1.0589 (k=3, m_0=10^{-2}) and* 1.3735 *at the E5 cell (d=64, N=16, $\rho^* = 0.1155$), where* (3) *gives $\rho_{\text{trap}} = 0.178$: $\rho_0 = 0.3$ is provably trapped, and $\rho_0 = 0.1 < \rho^*$ escapes. The escape half is proved only for the small-m system; for the exact flow it is a numerical statement.*

Remark 3 (N-flatness of the tied readout is proved)**.** *Part (iii)(c) closes the gap between the frozen-ρ identity (iii)(a) and the trap (iii)(b): for $\rho_0 \le \frac{1}{4}\rho^*(m_0)$ it bounds $\rho(t)$ on both sides up to $m = \frac{1}{2}$, and the tied escape time is $1/X$ up to a factor $1+O(\varepsilon+\rho_0+\lambda_k m_0^2)$ in which N enters only through $\varepsilon = k\rho_0/(Nm_0^{2k-2})$, so the N-flat third of the trichotomy is a theorem for $\varepsilon \ll 1$. When $\rho_0/(Nm_0^{2k-2}) = O(1)$ the readout can still decay substantially and the escape time is N-dependent: the E5 cell ($k=2$, $\rho_0 = 0.1$, $N=16$, $d=64$) has $\rho_0/(Nm_0^2) = 0.4$ and $\rho_0/\rho^*(m_0) = 0.87$ (outside the hypothesis); along the flow ρ falls to $0.72\rho_0$, and the cell showed an N-ratio of* 1.9 *(flow: $\tau(16)/\tau(256) =$ 1.86). `scripts/verify_tied_flat.py` integrates the exact flow on every tied cell of the paper:* 10 *of the* 19 *satisfy the hypothesis, and on them all bounds of (iii)(c) hold (*80/80 *checks; results/verify_tied_flat.txt).*

Theorem B (Free readout, fixed N, $d \to \infty$)**.** *Assume (B1) $\sigma = \sigma_k$, $k \ge 2$; (B2) $N \ge 2$, $r > 0$, $\Gamma_0 > 0$, $s > 0$ and $\mu_0 > 0$ fixed; (B3) $m_0 = \mu_0 d^{-1/2}$, $d \to \infty$. Then*

$$\tau_{\text{esc}} = (1+o(1))\,\exp\Big(\frac{(4k-2)\beta_2^2\Gamma_0^2}{2r}\Big)\,\frac{\mu_0^{-(4k-2)}\,d^{2k-1}}{4k(4k-2)\,\alpha_k^8\,s^2 N}, \qquad \beta_2^2 = 2k^2,\ \alpha_k = 1.$$

In particular the escape exponent is $4k$ for every finite N, and N enters only as the prefactor $1/N$.

Proof. Rescale $t \mapsto s^2 t$ (this maps the flow with s to the flow with $s=1$ and the same r), so $s = 1$. Put $m_1 = m_0 e^{-k^2\Gamma_0^2/r}$ and note $\frac{(4k-2)\beta_2^2\Gamma_0^2}{2r} = (4k-2)\frac{k^2\Gamma_0^2}{r}$, so the claim is $\tau_{\text{esc}} \sim \tau_B := m_1^{-(4k-2)}/(4k(4k-2)N)$. The equations are

$$\frac{d\ln m}{dt} = (1-m^2)\Big[4k\Gamma m^{2k-2}\big(1-\tfrac{\Gamma}{2}(1-\tfrac{1}{N})\big) - \frac{\Gamma^2 V'(m)}{Nm}\Big], \qquad \dot\Gamma = -2rD(m)\big(\Gamma - \Gamma^*(m)\big),$$

with $V'(m)/m = 4k^2(1+O(m^2))$, $D = N^{-1}(1+O(m^2))$, $\Gamma^* = Nm^{2k}(1+O(m^2+Nm^{2k}))$; constants below depend on $(k, N, r, \Gamma_0, \mu_0)$ only.

Phase 1, $0 \le t \le t_1 = \frac{(k-1)N}{4r}\ln d$ *(readout transient).* While $m \le 2m_0$, $\dot\Gamma = -\frac{2r}{N}(1+O(d^{-1}))\Gamma + O(d^{-k})$, and $\Gamma \ge \Gamma_0 d^{-(k-1)/2}/2 \gg Nd^{-k}$, so $\Gamma(t) = \Gamma_0 e^{-2rt/N}(1+o(1))$ uniformly. Integrating the m-equation, $\ln\frac{m(t_1)}{m_0} = -\frac{4k^2}{N}\int_0^{t_1}\Gamma^2(1+o(1)) + O\big(m_0^{2k-2}\int(\Gamma+\Gamma^2)\big) = -\frac{k^2\Gamma_0^2}{r} + o(1)$, because $\int_0^{t_1}\Gamma^2 = \frac{N\Gamma_0^2}{4r}(1+o(1))$ and $m_0^{2k-2}\int(\Gamma+\Gamma^2) = O(d^{1-k}N/r)$; the same bounds keep $m \le m_0(1+o(1))$, closing the bootstrap.

Phase 2, $t_1 \le t \le t_2 = t_1 + \frac{(k+1)N}{2r}\ln d$ *(relaxation onto* Γ^**).* Now $\Gamma \le 2\Gamma_0 d^{-(k-1)/2}$, so $|\frac{d\ln m}{dt}| = O(d^{1-k}\Gamma + \Gamma^2)$ integrates to $o(1)$: $m(t_2) = m_1(1+o(1))$. With $e = \Gamma - \Gamma^*$, $\dot e = -2rDe - \Gamma^{*\prime}\dot m$ and $D \ge 1/N$, $|e(t_2)| \le d^{-(k+1)}\Gamma_0 + \frac{N}{2r}\sup|\Gamma^{*\prime}\dot m| = o(d^{-k}) = o(\Gamma^*(m(t_2)))$.

Phase 3, adiabatic, until t_3*, the first time* $m = m_\delta$*.* Fix $\delta \in (0, \frac14)$ and choose m_δ (independent of d) so small that for $m \le m_\delta$ the $O(\cdot)$ terms above are $\le \delta$ and $\epsilon_\delta := 32k^2Nm_\delta^{4k-2} \le \delta r/N$. Write $\Gamma = \Gamma^*(1+e')$. While $|e'| \le 1$, $0 < \frac{d\ln m}{dt} = 4kNm^{4k-2}(1+e')(1+O(\delta))$ and, as $0 \le \frac{d\ln\Gamma^*}{d\ln m} = 2k - \frac{mD'}{D} \le 2k$, $|\frac{d\ln\Gamma^*}{dt}| \le \epsilon_\delta$; and $\dot e' = -\big(2rD + \frac{d\ln\Gamma^*}{dt}\big)e' - \frac{d\ln\Gamma^*}{dt}$. At $|e'| = \delta$ the right side has the sign of $-e'$, since $(\frac{2r}{N} - \epsilon_\delta)\delta > \epsilon_\delta$. Hence $|e'| \le \delta$ on $[t_2, t_3]$, and $t_3 - t_2 = (1+O(\delta))\frac{m(t_2)^{-(4k-2)} - m_\delta^{-(4k-2)}}{4k(4k-2)N} = (1+O(\delta)+o(1))\,\tau_B$.

Phase 4, $t \ge t_3$*.* $(m,\Gamma)(t_3) \in K_\delta = \{m_\delta\}\times[(1-\delta)\Gamma^*, (1+\delta)\Gamma^*]$, a compact set on which $L < 1 = L(0,0)$ (since $L = 1 - \Gamma G(2-(1+e'))$). L is a Lyapunov function, $\frac{dL}{dt} = -(1-m^2)(\partial_m L)^2 - r(\partial_\Gamma L)^2$, trajectories are bounded, and the only equilibria in $[0,1)\times[0,\infty)$ are $(0,0)$ and, for $m \in (0,1)$, points with $\Gamma = \Gamma^*(m)$ and $\frac{d}{dm}L(m, \Gamma^*(m)) = -R'(m) = 0$ (envelope theorem), which Lemma A.1 excludes. By LaSalle every trajectory from K_δ has $m \to 1$; the sets $\{x :\ m \text{ exceeds } \frac12 \text{ before time } T\}$ are open and cover K_δ, so the remaining time is $\le T_\delta < \infty$, independent of d.

Phases 1, 2 and 4 last $O(N\ln d/r) + T_\delta = o(\tau_B)$, as $\tau_B \asymp d^{2k-1}$. Thus $\limsup|\tau_{\rm esc}/\tau_B - 1| \le C\delta$ for every δ. □

Proposition 4 (Joint scaling; proof sketch)**.** *Let* $N = \lambda d^k$*,* $r = \tilde r d$*,* $m_0 = \mu_0 d^{-1/2}$*,* $\Gamma_0 \in [0,2)$*,* $s = 1$*, and put* $\mu = \sqrt{d}\,m$*,* $\theta = t\,d^{1-k}$*. As* $d \to \infty$ *the flow converges, uniformly on compacts of* $\{\mu > 0\}$ *with an* $O(1/d)$ *error, to the d-free system*

$$\frac{d\mu}{d\theta} = 4k\,\Gamma\big(1 - \tfrac{\Gamma}{2}\big)\mu^{2k-1}, \qquad \frac{d\Gamma}{d\theta} = 2\tilde r\Big[\mu^{2k} - \Gamma\Big(\mu^{2k} + \frac{1}{\lambda}\Big)\Big], \tag{7}$$

whose solution blows up ($\mu \to \infty$) *at a finite* $T(\mu_0; \lambda, \tilde r, \Gamma_0)$*, and* $\tau_{\rm esc}\,d^{1-k} \to T$*. (The finite-N repulsion is* $O(1/d)$ *in this scaling.) With* $\kappa := 2 - \partial\ln T/\partial\ln\mu_0$ *(the secant exponent of the experiments is a finite-difference estimate of it): (a)* $\lambda T \to \mu_0^{-(4k-2)}/(4k(4k-2))$ *as* $\lambda \to 0$ *(*$\tilde r, \Gamma_0, \mu_0$ *fixed):* $\kappa \to 4k$*; (b) at* $\lambda = \infty$ *and fixed* $\Gamma_0 \in (0,2)$*,* $\mu_0^{2k-2}T \to 1/\big((2k-2)\,2k\Gamma_0(2-\Gamma_0)\big)$ *as* $\mu_0 \to 0$*:* $\kappa \to 2k$*; (c) at* $\lambda = \infty$*,* $\Gamma_0 = 0$ *(readout growth-limited),* $\mu_0^{2k-1}T \to \frac{\pi}{2}\frac{(2k-3)!!}{(2k-2)!!}\big/\big(2\sqrt{2k\tilde r}\big)$ *as* $\sqrt{\tilde r}\mu_0 \to 0$*:* $\kappa \to 2k+1$*.*

Sketch. Substituting, $d^{k-\frac12}\dot m = (1-\frac{\mu^2}{d})\big[2k\Gamma(2-\Gamma(1-\frac{1}{N}))\mu^{2k-1} - \frac{\Gamma^2}{\lambda}\sum_i 2i\beta_{2i}^2\mu^{2i-1}d^{-i}\big]$ and $d^{k-1}\dot\Gamma = 2\tilde r[\mu^{2k} - \Gamma((1-\frac1N)\mu^{2k} + V(\mu d^{-1/2})/\lambda)]$, which is (7) up to $O((1+\mu^{2k+2})/d)$. In (7), $\Gamma \le \max(\Gamma_0, 1) < 2$ (at $\Gamma \ge 1$, $\dot\Gamma < 0$) and $\Gamma \ge c := \min(\Gamma_0, \Gamma^*_\infty(\mu_0))$ with $\Gamma^*_\infty = \mu^{2k}/(\mu^{2k} + \lambda^{-1})$ increasing along the (increasing) μ (for $\Gamma_0 = 0$, $\Gamma > 0$ for all $\theta > 0$ since $\dot\Gamma = 2\tilde r\mu^{2k} > 0$ there); so $\dot\mu \ge c'\mu^{2k-1}$ and μ blows up in finite time. Split at $\mu = M$: up to M, continuous dependence gives convergence of the hitting time; after M both the limit and (by the same invariance bounds, the repulsion being $O(M^{2-2k}/\lambda d)$) the finite-d flow need time $\le CM^{2-2k}$, which $\to 0$ as $M \to \infty$. Corners: (a) Γ relaxes at rate $2\tilde r/\lambda \to \infty$ onto $\Gamma^*_\infty \approx \lambda\mu^{2k}$, the transient moves $\ln\mu$ by $O(\Gamma_0\lambda/\tilde r)$, and

$\dot\mu = 4k\lambda\mu^{4k-1}(1+o(1))$. (b) $d\Gamma/d\mu = \tilde r\mu(1-\Gamma)/(2k\Gamma(1-\Gamma/2))$, so $\Gamma = \Gamma_0 + O(\tilde r\mu_0^2 M^2)$ on $[\mu_0, M\mu_0]$, which carries all but $O(M^{2-2k})$ of T. (c) For $\Gamma \ll 1$, $\Gamma^2 = \frac{\tilde r}{2k}(\mu^2-\mu_0^2)$ is an approximate first integral, so $\dot\mu = 4k\sqrt{\tilde r/2k}\,\sqrt{\mu^2-\mu_0^2}\,\mu^{2k-1}$ and $T = \mu_0^{1-2k}\int_1^\infty \frac{dx}{4k\sqrt{\tilde r/2k}\,x^{2k-1}\sqrt{x^2-1}}$; the region where $\Gamma = O(1)$ costs $O(\tilde r^{k-1}) = o(\mu_0^{1-2k}\tilde r^{-1/2})$. The interchange of the corner limits with $\partial/\partial\ln\mu_0$ (needed for κ rather than for T) is not proved here; numerically the limits hold for the derivative as well (script). □

Remark 4 (What is needed to transfer Theorem B to online SGD; heuristic)**.** *(1)* Feature step. *A drift $\eta\, c\, m^{\kappa-1}$ beats the spherical renormalisation drag $\asymp \eta^2 d\,\sigma_w^2 m/B$ at $m \asymp d^{-1/2}$ iff $\eta \lesssim cB\,d^{-\kappa/2}/\sigma_w^2$: the critical scaling $\eta \asymp d^{-\kappa/2}$ of Ben Arous et al. [2], with $\kappa = 2k$ (pinned, tied) or $4k$ (free). Every w-gradient carries a factor Γ, so $\sigma_w^2 \asymp \Gamma^2/N$ near $m = 0$, which relaxes the free-readout requirement once $\Gamma \approx \Gamma^*$. (2)* Readout noise. *The SGD readout fluctuates around Γ^* with stationary variance $\mathrm{Var}(\Gamma) \asymp \eta_\Gamma/B$ (relaxation rate and per-prompt variance both $\asymp 1/N$), independent of N; as L is quadratic in Γ this adds $\mathrm{Var}(\Gamma)D'(m)$ to the repulsion. Pathwise tracking, $\mathrm{Var}(\Gamma) \ll \Gamma^{*2} \asymp N^2 d^{-2k}$ at $m \asymp d^{-1/2}$, needs $\eta_\Gamma \ll BN^2 d^{-2k}$; the mean m-drift alone needs only $\eta_\Gamma \ll BN^2 d^{1-2k}/k$. At $\eta_\Gamma = r\eta_0 d^{-2}$ and fixed N both fail as $d \to \infty$ ($k \geq 2$). In E2 ($k = 2$, $d = 64$, $\eta_0 = r = 1$) $d^2 = 4096$ against $BN^2 \geq 16384$; the E5 cell $\eta_\Gamma = 10\eta$, $N = 16$ violates it ($BN^2/r = 1638 < d^2$, constants aside), consistent with SGD being 24% slower than the flow there (2.26M vs 1.81M steps).*

Conjecture 1 (SGD transfer)**.** *(a) Under the conditions of Theorem B, with $\eta = \eta_0 d^{-2k}$, $\eta_\Gamma = r\eta$ and $\eta_\Gamma \ll BN^2 d^{-2k}$ (i.e. $B \gg r\eta_0/N^2$), $\eta \cdot T_{0.5}/\tau_{\mathrm{esc}} \to 1$ in probability. (b) In the setting of Theorem A(i) with $m_0 \asymp d^{-1/2} < m^*$, for every B and every $\eta \leq \eta_0 d^{-k}$, $\Pr\big(\max_{n \leq e^{c\,d\,m^{*2}}} m_n \geq m^*/2\big) \to 0$; this would follow from a supermartingale bound for $e^{a\,d\,m^2}$ on $\{m \leq m^*\}$ (negative population drift plus spherical drag), in the manner of Ben Arous et al. [2].*

C Experimental details

All toy experiments: float64 numpy, analytic gradients verified by finite differences (relative error $< 10^{-4}$), online spherical SGD with fresh data each step, $B = 32$ prompts per step unless stated, $\eta = \eta_0/d^2$. Projected samplers (exact in distribution) were used where the gradient depends on x only through a few projections. Compute: Exp. 1 6.4, Exp. 1b 2.8, Exp. 2 3.2, Exp. 3 2.3, Exp. 4 6.0, Exp. 5 6.0 CPU-hours on 4 cores.

Exp. 1 (single skill, uniform initialisation). $k^* \in \{1,2\}$, $d \in \{8,\dots,128\}$, 5 seeds, $N = 128$, fixed $\gamma = 0.1$ (A) and $a = 1$ (B). Slopes of $\log T_{0.5}$ vs $\log d$: B $k^* = 1$: 1.98 ± 0.06 ($d \geq 16$; the pre-registered 1.5 was an error: for $\kappa < 2$ the drift is $O(1)$ and $T \propto 1/\eta = d^2$); B $k^* = 2$: 2.33 ± 0.22; A $k^* = 1$: 2.71 ± 0.43; A $k^* = 2$: 3.1–3.5 (20 of 105 runs censored at $3 \cdot 10^6$ steps). Regression of $\log(T_{0.5}\eta)$ on $\log|m_0|$: A $k^* = 2$: -2.19 ± 0.08 (reached runs with $\eta_0 \in \{1,2\}$, $d \geq 16$; -2.29 ± 0.07 on all reached runs), i.e. $\kappa \approx 4.2$–4.3, a lower bound because of censoring; B $k^* = 2$: -0.57 ± 0.06 (logarithmic, $\kappa = 2$); A $k^* = 1$: -0.50 ± 0.06; B $k^* = 1$: -0.13 ± 0.02.

Exp. 1b (fixed m_0). Threshold grid: $\gamma \in \{1, 0.1\}$, $N \in \{32, 128, 512\}$, $d \in \{16, 32, 64\}$, 3 seeds, $3 \cdot 10^5$ steps. Escapes at $\gamma = 1$: $N = 32$: $2/3, 0/3, 0/3$; $N = 128$: $3/3, 1/3, 0/3$; $N = 512$: $3/3, 3/3, 3/3$ (corrected $d^* = 6.8$, 30.8, 126.8). $\gamma = 0.1$: 27/27. P7: $\gamma = 1$, $N = 128$, $d = 64$, $B \in \{32, 256, 1024\}$ with $3 \cdot 10^5, 10^5, 3 \cdot 10^4$ steps: 0/9 escapes, final $|m|$ 0.005–0.015 ($B \leq 256$) and 0.063–0.067 ($B = 1024$). Exponent table ($d = 32$, 4 values of m_0, 3 seeds, $\eta = 1/d^2$, $\Gamma_0 = 0.01$ or

$\rho_0 = 0.01$): in Fig. 1; censored runs ($N = 32$: 3/12 for both free protocols; $N = 128$, 10η: 2/12) are all at the smallest m_0.

Censored fits and absolute-time ratios. For the three E1 cells containing censored runs (all at the smallest m_0) we refit $\log(T_{0.5}\eta) = a + b \log m_0$ by a Tobit likelihood with right-censoring at the step cap ('results/README_stats.md'): κ_{eff} becomes 6.05 ± 0.18 (free $\eta_\Gamma = \eta$, $N = 32$; ODE 6.47), 6.73 ± 0.18 (free 10η, $N = 32$; ODE 7.02) and 6.67 ± 0.37 (free 10η, $N = 128$; ODE 6.18); the other nine cells are unchanged. All twelve are within 0.49 of the ODE and ten within 2 s.e.; bootstrap s.e. in the censored cells are 0.21–0.49. The Tobit fit assumes censored runs would eventually cross $m = \frac{1}{2}$, which is an assumption (their final $|m|$ is 0.04–0.11). Absolute-time ratios median($T_{0.5}$)/ODE: E1 (45 reached cells, ODE at each run's exact m_0): 24/35/45 within 10/20/50%, range 0.64–1.34; E2 (18 cells, pinned $\gamma = 1$ excluded as the ODE predicts a trap): 17/18/18, range 0.85–1.10. Single seeds range 0.50–2.08×. One censored E1 cell (free 10η, $N = 128$, $m_0 = 0.5d^{-1/2}$: 1/3 reached) is at least 2.35× slower than the ODE.

Exp. 6 (three regimes). $d = 64$, $m_0 = d^{-1/2}$, $\eta = 1/d^2$, $N \in \{16, 64, 256\}$, schemes $NB = 4096$ and $B = 64$, protocols pinned $\gamma \in \{1, 0.1\}$, free ($\Gamma_0 = 0.01$, $\eta_\Gamma = \eta$), tied ($\rho_0 = 0.01$), 3 seeds, 10^6 steps ($2.5 \cdot 10^6$ for the free $N = 16$ rerun). ODE predictions were computed and committed before the run. 12 of 72 runs are duplicates ($N = 64$ has $B = 64$ in both schemes). Compute 2.4 CPU-h. Medians of three seeds resolve escape times to roughly 5%. ODE values that moved by more than 1% between the Euler and LSODA integrators were updated to LSODA ('scripts/ode_lsoda.py'); values within 1% are quoted as originally computed; the SGD/ODE ratio statistics above were computed with the earlier Euler values and would rise by 0.6–3.6%.

Exp. 7 (controls, E5). As Exp. 6 with $B = 64$, 3 seeds, cap $2.5 \cdot 10^6$ (10^6 for the tied $\rho_0 = 0.3$ cell): free $\Gamma_0 = 0.01$, $\eta_\Gamma = 10\eta$: 2.26M/498k/212k (ODE 1.81M/500k/205k); free $\Gamma_0 = 0.1$, $\eta_\Gamma = \eta$: 1.52M/271k/166k (ODE 1.49M/270k/161k); tied $\rho_0 = 0.1$: 37k/18k/17k (ODE 33k/19k/17k; one $N = 16$ seed at 567k); tied $\rho_0 = 0.3$: 716k/27k/21k (ODE: trap/31k/21k); pinned $\gamma = 0.01$: 1.71M/1.64M/1.62M (ODE 1.70M/1.64M/1.63M); $\eta \propto B$ at pinned $\gamma = 0.1$, $N = 64$: 773k/192k/48k steps for $B = 16/64/256$, flow time 47.2/47.0/46.9. Compute 2.2 CPU-h.

Exp. 2 (many skills). $P = 16$, $\pi_p \propto p^{-1.5}$, $M = 64$, $\Gamma = 0.1$, $N = 128$, 3 seeds (plus 2 extra and $\alpha = 1$). Skills 1–8 learned in every run; emergence of skills 9–16 partial. In-weight baselines: (i) readout $\rho = \sum_p a_p/M$ (the readout-1 baseline does not learn); (ii) the 2-homogeneous student $\sum_j \|u_j\|^2 \sigma_2(\hat{u}_j \cdot x)$, $\|u_j\|_0^2 = 0.01$, $\eta \in \{0.25, 1, 4\}/d^2$, 3 seeds each: slope 0.75 ± 0.09 (1000-step log), 0.87–0.89 ± 0.09 (100-step log), $T_p \eta a_p = 0.23$–0.26; $18.7, 10.7, 9.0, 6.3, 2.7, 3.0, 2.7, 3.7$ features on skills 1–8, best single alignment $0.86 \to 0.59$, pooled alignment 1.000 for all 16 skills, population MSE $3 \cdot 10^{-8}$: collective fit. Both are collective at $M = 64 > d$; a decoupled in-weight regime would need $M < d$ (as in Ben Arous et al. [3], whose plateau length $\log(d/r_s)$ comes from the Gram matrix $WW^\top$ having eigenvalues $\approx r_s/d$ at initialisation—at $M \geq d$ that matrix is already full rank, which is our reading of why no search plateau and no single-feature emergence appear) and was not run. Slopes and exponents as in Sec. 4; the $\alpha = 1$ slope is 0.88 ± 0.09 (A) vs 0.61 ± 0.08 (B).

Exp. 3 (composition). $P = 2$, $\pi = (0.75, 0.25)$, $M = 4$, $\Gamma = 0.1$, $N = 128$, 5 pre-registered seeds + 5 extra + 3 at $\Gamma = 0.3$. Additivity residual with a paired estimator (same inputs, 17 queries per context): $\max |R| \leq 0.0086$ over 13 runs (the pre-registered independent-set estimator was too

noisy, 0.04–0.35). Final neuron assignments (skill 1, skill 2): $(4,0),(2,2),(2,2),(2,1),(2,2)$ for the pre-registered seeds. Neuron arrival times $T \approx K/(\eta\pi_p m_0^2)$ with $K = 0.60$ (0.54–0.78).

Exps. 4–5 (transformers). Tokens $[x_i; y_i]$, query $[x_q; 0]$; embedding Linear(d+1 $\to$ 64), 2 pre-LN blocks (4-head causal softmax attention + ReLU MLP), linear readout; Adam 10^{-3}, fp32, online data, eval every 200 steps on 2048 held-out prompts, T_e = first eval with normalised MSE < 0.5. Exp. 4: MLP width 256, $d \in \{16, 32\}$, $N \cdot B = 8192$. Exp. 5: $d = 256$, width 32, cells (a) N=256, B=64: $T_e = 4000, 2800$, third seed censored at 17k on the plateau; (c) N=16, B=64: none in 30k (3/3); (d) N=16, B=1024: 14800, 10600, 11600; (e) $k^* = 1$, N=16, B=64: 1400, 1200; (f) in-weight: 400, 200; initial $\max_j |\cos(W_j, v)|$: 0.126, 0.152, 0.176. Exp. 9 ($k^* = 3$, single neuron, `sweep9.py`): $\eta = 2 \cdot 10^{-4}$ for all d, $B = 64$, $m_0 = d^{-1/2}$ exactly, cap 10^6 steps, 3 seeds, 27 runs, 0.89 CPU-h; medians/ODE tied N=128: 1.18, 0.89, 0.80 ($d = 8, 16, 32$), pinned γ=0.2 d=8: 1.02; post-hoc controls: d=8 at $\eta/4$: 1.007; d=32 at B=256: 1.14 (declared before running). Exp. 8 (`transformer_exp8.py`, same model/data code as Exp. 5): readout frozen at its initial norm (0.56–0.58) or at 0.1× it, or trainable; cells N=16, B=64; N=16, B=1024; N=256, B=64; 2 seeds; 30k steps; 5.5 CPU-h; one pinned-small N=256 seed censored at 17k (budget). Exp. 12 (`ablation_exp12.py`, torch): single feature, σ_2, d=64, pinned Γ=1, {linear, softmax}×{SGD η=$1/d^2$, Adam 10^{-3}}×{N=16, B=64; N=16, B=1024; N=1024, B=64}, 2 seeds, caps $3 \cdot 10^5/10^5$ steps, 3.1 CPU-h; escapes only in linear N=1024 (SGD 34.5k/37.4k, Adam 379/1421). Exp. 13 (`ablation_exp13.py`): as Exp. 12 with the distance-based softmax, β=0.3, full data, caps $5 \cdot 10^5$ (SGD) / 10^5 (Adam), 12 runs, 2.7 CPU-h; Monte-Carlo drift table `results/exp13/drift_mc.txt`. Exp. 14 (`ablation_exp14.py`): Exp. 13's softmax+Adam cell at N=16, $B \in \{16, 64, 256, 1024\}$, 3 seeds, 0.15 CPU-h. Exp. 15 (`ablation_exp15.py`): linear single feature, pinned Γ=1, N=1024, $B \in \{16, 64, 256, 1024\}$, 3 seeds; Adam 10^{-3} (full data, cap $2 \cdot 10^4$) and SGD η=$1/d^2$ (projected sampler, cap $3 \cdot 10^5$); 1.4 CPU-h. Exps. 16–17 (`transformer_exp8.py --run free 16 B seed`): N=16, $B \in \{256, 512, 2048\}$ at 30k steps (5 runs, 3.3 CPU-h) and B=256 at 80k (2 seeds, 2.5 CPU-h); $T_e(0.5)$: 256: stuck/stuck (80k); 512: 13400/stuck; 1024: 14800/10600 (Exp. 5); 2048: 11000. Exp. 18 (`transformer_exp18.py --variant D1|NL`): one block / no LayerNorm, cells N=256, B=64; N=16, B=1024; N=16, B=64, 2 seeds (one dropped), 6.0 CPU-h; $T_e(0.5)$ D1: cens.18.6k/2400; 24600/stuck; stuck/stuck. NL: cens.10.8k/1600; stuck/13200; stuck/–. Exp. 19 (`transformer_exp8.py --run free 16 B seed`, cap 40k): $B \in \{384, 512, 768\}$, seeds 0–2, 5.6 CPU-h; $T_e(0.5)$: 384: 17800/stuck/stuck; 512: 13400/stuck/12400; 768: 11000/stuck/4400. Exp. 11 (`transformer_exp11.py`): `attn.out` of both blocks frozen at init (norms 4.6) or at 0.1× (norms 0.46), same cells, 2 seeds, 30k steps, 6.0 CPU-h; one 0.1× cell has one seed, two 0.1× runs censored on the plateau at 8k and 10.8k (budget). Exp. 10 (`sweep10.py`): E2's protocols at $d = 128$, $\eta = 1/d^2$, $B = 64$; pinned γ=1 (2 seeds, cap 10^6), pinned γ=0.1 (3 seeds, cap $2.5 \cdot 10^6$), tied ρ_0=0.01 (3 seeds), free Γ_0=0.01, η_Γ=η at N=256 (2 seeds); 26 runs, 3.8 CPU-h.

Ablation ladder (Exps. 12–14) and the plateau. On the transformer's plateau the final readout norm collapses from 0.57 to 0.03–0.16 in stuck and emerging runs alike, and in one of two seeds the first norm logged after emergence is back at 1.7 (the other ends at 0.08); the collapse is therefore plateau behaviour, not a signature of Prop. 3. Exp. 12 (single feature, pinned $\Gamma = 1$, $d = 64$, $N \in \{16, 1024\}$, $B \in \{64, 1024\}$, two seeds): with the linear statistic, Adam (10^{-3}) leaves $N = 16$ stuck at both B (0/4) and escapes $N = 1024$ in 379/1421 steps (P25a–b held); a product-logit softmax statistic $\sum_i \mathrm{softmax}_i(\sigma(w\cdot x_i)\sigma(w\cdot x_q))y_i$ has negative population drift everywhere we checked and did not learn at any N (P25c–d failed as worded; inconclusive). Exp. 13 (distance-based softmax $\sum_i \mathrm{softmax}_i(-0.3(\sigma(w\cdot x_i) - \sigma(w\cdot x_q))^2)y_i$): under SGD $N = 16$ is stuck 0/4 at

$5 \cdot 10^5$ steps and $N = 1024$ escapes (63.9k/65.8k; P26a held); under Adam $N = 16$ escapes in 3/4 runs already at $B = 64$ (26k, 25k; P26b failed). The honest reading is about barrier depth, not an interaction: the distance-based statistic's population drift at m_0, $N = 16$ is $-6 \cdot 10^{-3} \pm 6 \cdot 10^{-3}$ (indistinguishable from zero), so Adam crosses a barrier that may not exist, while it does not cross the linear model's deep one ($m^* = 0.60$). Exp. 14 (that cell, $B \in \{16, 64, 256, 1024\}$, three seeds) is consistent with a noise-driven escape: escape fraction $3/3, 3/3, 3/3, 2/3$ and median time 8.0k/25k/39k/60k, rising with B (P27 held on the literal reading of its monotonicity clause and fails on a strict "one tie" reading, since the fraction saturates at 3/3 twice); the seed ranges overlap heavily ($B = 16$: 5.5–35k; $B = 1024$ contains a 3.6k run). None of these cells reproduces the transformer's pattern (stuck at $B = 64$, rescued at $B \geq 512$–1024).

D Pre-registration summary

All predictions were written to a version-controlled document before the corresponding run; the full text with timestamps, outcomes and changelog is in the repository (`docs/preregistration.md`). Summary:

ID	prediction	outcome
P1	MC drift of A, k^*=2, $\propto m^3$; closed form within 20%	met (slope 3.14 ± 0.07; $\leq 3\%$ at $m \geq 0.2$)
P2	γ=1, N=128: stuck for $d \geq 32$	partly (d=64 stuck, d=32: 1/3 escaped); threshold formula later corrected, $d^* \approx 31$
P3	γ=0.1: escapes at all d	met (27/27)
P4	free readout: κ rises as N falls, $4 \to 8$	direction met; magnitudes match the ODE at d=32 ($4.85 \to 7.0$), not the asymptotes
P4′	tied readout: $\kappa \approx 4$ at all N; ρ_0=1 stuck for $d \geq 32$ (flow traps; SGD delays)	κ met (4.10–4.44); ρ_0=1: d=32 escaped 3/3, d=64 stuck
P5	learned $\Gamma \approx \Gamma^*(0.5; N)$	not met (relaxation too slow at $\eta_\Gamma = \eta$)
P6	A vs B slope difference ≈ 1 at k^*=2	met via m_0-regression (κ 4.2 vs 2); d-slopes 0.9 ± 0.5
P7	no escape at any B	met (0/9)
P8	transformer, d=32: T_e falls $\geq 3\times$ from N=16 to 256 at fixed tokens	**failed, reversed** ($1000 \to 2800$)
P9	k^*=1 weak N-dependence; in-weight fastest	partly ($\times 2.0$; in-weight fastest)
P10	N-dependence in both softmax and linear attention	met (same direction)
P11a–b	d=256, width 32: (c) stuck, (d) stuck	(c) stuck 3/3; (d) emerged 3/3 at $\approx 3\times$ (a): **both rival and default failed**
P11c	k^*=1 and in-weight controls fast	met
P12	additivity residual $\lvert R \rvert \leq 0.02$	failed as pre-registered (estimator noise); identity confirmed with a paired estimator
P13	no extra compositional delay	supported with substitute statistic (floor made the original unreachable)
P14	multiplicative accuracy, Gaussian closed form	approximately multiplicative; Gaussian form wrong
P15	product composition not learned	met
P19	d=64: pinned-1 trapped; free $T \propto$ ODE with $T(16)/T(256) = 4.4$; tied flat within 5%	pinned-1: 0/9 at $N \leq 64$, 3/6 at N=256; free within one s.e. of ODE, ratio 4.4; tied flat within $\times 1.25$ (5% not resolvable)
P20	controls: ratios 8.9/9.1 (free $10\eta/\Gamma_0$=0.1), tied ρ_0=0.1 ratio 1.9, tied ρ_0=0.3 traps at N=16, pinned 0.01 flat, $\eta \propto B$ flow-invariant	10.7/9.2; 2.15; **escaped after** $30\times$ **delay (failed as worded)**; 1.06; invariant to 0.5%
P21	transformer readout protocols: pinned O(1) $\Rightarrow$ N=16 stuck at B=1024; pinned small $\Rightarrow$ N=16, B=64 emerges; free reproduces Exp. 5	**failed** (8400, 5800 at B=1024); **failed** (stuck 2/2); held ($3.7\times$, $3.8\times$)—kill criterion met
P24	transformer attention-output scale: frozen O(1) $\Rightarrow$ N=16 stuck at B=1024; frozen $0.1\times$ $\Rightarrow$ N=16, B=64 emerges; O(1) within $1.5\times$ of free at N=256	**failed** (15.0k, 26.4k); **failed** (stuck 2/2); **failed** ($3.4\times$, $2.6\times$ slower)—kill criterion met
P25	single-feature ablation: Adam and softmax each keep N=16 stuck at B=64 and 1024 while N=1024 escapes	held (linear+SGD); held (linear+Adam: 0/4 stuck, N=1024 2/2); **failed as worded** (softmax: N=1024 also stuck 0/2, inconclusive) $\times 2$
P26	distance-based softmax (β=0.3): N=16 stuck at both B, N=1024 escapes, under SGD and Adam	held (SGD: 0/4; 63.9k/65.8k); **failed** (Adam: N=16 escapes 3/4, B=64 26k/25k)
P27	softmax+Adam escape at N=16 is noise-driven: escape fraction non-increasing and median $T_{0.5}$ non-decreasing in $B \in \{16, 64, 256, 1024\}$	held on the literal reading (3/3, 3/3, 3/3, 2/3; medians 8.0k/25k/39k/60k); fails a strict "one tie" reading (fraction saturates twice)
P28	Adam makes B matter where the drift is positive (linear, N=1024): slope of $\log T$ vs $\log B$ in $[-0.75, -0.25]$; SGD flow time B-invariant	**failed as worded** (Adam -0.754 ± 0.144, 0.004 outside; $T(16)/T(1024) = 17$); **failed on spread** (SGD slope -0.015, medians span 1.20 vs 15%)
P29–P30	transformer B-scan at N=16: graded B-dependence (B=256, 512 emerge; slope in $[-1.2, -0.3]$); B=256 emerges within 80k	**failed** (B=256 stuck 2/2 at 30k; 512: 1/2; slope -0.51 on censored medians); **failed** (B=256 stuck through 80k/77k)
P31	transformer without the second block (D1) / without LayerNorm (NL): pre-interpreted patterns A/B/C/0; weak priors D1→B or 0, NL→A	both Pattern A (control escapes; N=16, B=64 stuck; N=16, B=1024 1/2); D1 prior missed, NL prior matched
P32	transformer B-scan at N=16, 3 seeds, $B \in$	**neither** (1/3, 2/3, 2/3, 3/3; slope $-1.15 \pm$